

Write-up on Hasse Weil Functions

Chunye Yang

October 7, 2026

This will be a write-up on some algebraic side of the story of the Hasse-Weil Zeta function along side what will be discussed in Koblitz's book. We will mainly follow [Mus11] which is a lecture notes at the University of Michigan.

Since the writing of the original lecture notes are already very good, in this write-up, we will try to fill in as much details as possible. If you feel you are now seeing the main idea by the amount of details here, go to the original lecture notes in [Mus11].

1 Basics of Hasse-Weil Zeta Functions

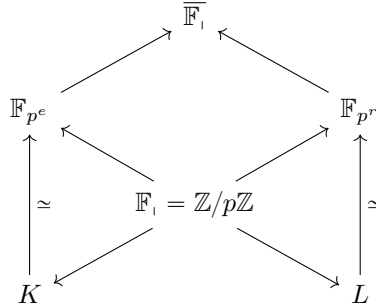
1.1 Review on Finite Fields

Recall that if k is a finite field, then $|k| = p^e$ for some $e \geq 1$ where $p = \text{char } k$. Further, two finite fields L, K of the same cardinality are isomorphic. Indeed, whose fields have a common subfield $\mathbb{Z}/p\mathbb{Z}$ and we know that they are both the splitting field of $x^{p^e-1} - 1$, therefore the isomorphism of $\mathbb{F}_p$ -algebras can be defined via the fact that they are both isomorphic to $\mathbb{F}[x]/(x^{p^e-1} - 1)$ and we know that for example the isomorphism can be defined by:

$$\mathbb{F}_p[x]/(x^{p^e-1} - 1) \rightarrow K, \quad x \mapsto y$$

where y is the generator of the cyclic subgroup K^* . We will denote a finite field with $q = p^e$ elements $\mathbb{F}_q$. If we are using $\mathbb{F}_p$ we are saying that this is the field $\mathbb{Z}/p\mathbb{Z}$.

Recall that for any $\mathbb{F}_q$, it contains $\mathbb{F}_p$, therefore we know that we can embed it to an algebraic closure $\overline{\mathbb{F}_p}$ of $\mathbb{F}_p$. However, I claim that in a fixed algebraic closure of $\mathbb{F}_p$, there is only a unique subfield with q elements. Indeed, this subfield should be the elements in $\overline{\mathbb{F}_p}$ such that $x^q = x$. Therefore the image of $\mathbb{F}_q$ is just this unique subfield. Then in the future, if we say $\mathbb{F}_{p^r}$ we are saying that we work in an algebraic closure of $\mathbb{F}_p$.



From the diagram you can also say that an $\mathbb{F}_p$ -algebra map between K, L is equivalent to such a map between $\mathbb{F}_{p^e}$ and $\mathbb{F}_{p^r}$.

One consequence of the above discussion is that if $\mathbb{F}_{p^e}$ and $\mathbb{F}_{p^r}$ are two finite fields, then there is a morphism of $\mathbb{F}_p$ -algebra from $\mathbb{F}_{p^e}$ to $\mathbb{F}_{p^r}$ iff $e \mid r$. Indeed, if $e \mid r$, we can first embed them into $\overline{\mathbb{F}_p}$ and then in this algebraic closure, we know that if $e \mid r$, i.e. $r = ne$, then if $x^{p^e} = x$, then we know that $x^{p^r} = x^{p^{ne}} = x^{(p^e)^n} = ((x^{p^e})^{p^e} \dots)^{p^e} = x$, thus x is also in the subfield determined by $x^{p^r} = x$. In particular, two finite fields with p^r, p^e elements are isomorphic iff $e = r$.

Notation 1.1. From now on, we will fix $k = \mathbb{F}_q$ for some $q = p^e$.

Then we can also take the algebraic closure of $\mathbb{F}_p$ as the algebraic closure of k , which we denote by $\bar{k}$. Then consider $\text{Gal}(\bar{k}/k)$, we know that there is an special element σ which is defined by $x \mapsto x^q$. This is called the **arithmetic Frobenius element**. It's inverse in $\text{Gal}(\bar{k}/k)$ is called the **geometric Frobenius element**. It's called the geometric Frobenius because from the geometric or cohomological side, the inverse of σ is more natural than σ .

Again, there is a unique subextension K of k in $\bar{k}$ of degree r and you can check that it's the subfield of $\bar{k}$ fixed by σ^r . Moreover, $\text{Gal}(K/k)$ is cyclic generated by $\sigma|_K$.

Finally, σ is the topological generator of $\text{Gal}(\bar{k}/k)$.

1.2 Varieties over Finite Fields

By a variety over k , we simply mean a reduced and finite type scheme over k with no separability or irreducibility assumed. If we need those assumptions later, we will make it clear.

As a topological space, we will use X_{cl} to denote the set of closed points of X . Given such a point $x \in X_{\text{cl}}$, we can consider the local ring $\mathcal{O}_{X,x}$ and the residue class field $k(x)$. Since X is finite type over k , we know that $k(x)/k$ is a finite extension by Hilbert's Nullstellensatz. We will put $\deg(x) = [k(x) : k]$.

Besides the closed points, suppose that $k \rightarrow K$ is any field extension, we can consider the K -values points, which are defined to be:

$$X(K) = \text{Hom}_k(\text{Spec} K, X)$$

Notice that given a k -morphism $\text{Spec} K \rightarrow X$, we can first consider the unique point x of the image and we know that we have the local ring map induced by the morphism of schemes (which is also a k -algebra map):

$$\mathcal{O}_{X,x} \rightarrow K$$

This gives a unique k -algebra map $k(x) \rightarrow K$ by quotient out the maximal ideal of $\mathcal{O}_{X,x}$.

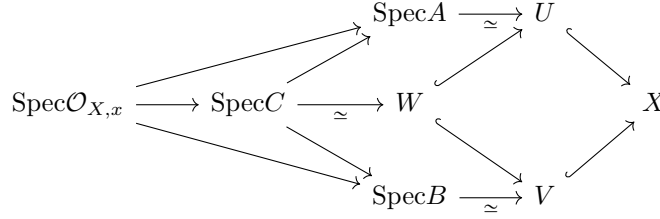
Conversely, given a k -algebra map $k(x) \rightarrow K$ for some $x \in X$, we consider the composition:

$$\mathcal{O}_{X,x} \rightarrow k(x) \rightarrow K$$

which gives a unique morphism of k -schemes $\text{Spec} K \rightarrow \text{Spec} \mathcal{O}_{X,x}$. Recall that there is a canonical map of schemes $\text{Spec} \mathcal{O}_{X,x} \rightarrow X$ such that the unique maximal ideal of $\mathcal{O}_{X,x}$ is mapped to x . Indeed, this mapped is defined by first choosing an affine open neighborhood U of x . Say it's isomorphic as an k -scheme to $\text{Spec} A$ where A is a finitely generated k -algebra, then this isomorphism maps $x \in U$ to the prime $p \in \text{Spec} A$, which induces an isomorphism of k -algebras $A_p/pA_p \simeq k(x)$. Then we can define the morphism $\text{Spec} \mathcal{O}_{X,x} \rightarrow X$ by the following diagram:

$$\begin{array}{ccccc} \text{Spec} A & \xrightarrow{\simeq} & U & \hookrightarrow & X \\ \uparrow & \searrow & \swarrow & & \\ \text{Spec} \mathcal{O}_{X,x} & \longrightarrow & \text{Spec} k & & \end{array}$$

where the morphism $\text{Spec} \mathcal{O}_{X,x} \rightarrow \text{Spec} A$ is defined by the map $A \rightarrow A_p/pA_p \simeq k(x)$. Then we can check that this definition doesn't depend on the choice of the affine open we choose. This can be done since if you choose another affine open $V \simeq \text{Spec} B$ containing x . Then we can choose another affine open $\text{Spec} C$ in $U \cap V$ and show that the map defined by U, V is the same as the map defined using $U \cap V$, which is given in the following diagram:



Since open embeddings $U \hookrightarrow X$ and $V \hookrightarrow X$ are monomorphisms. We only need to check that the triangle consisting of $\text{Spec } \mathcal{O}_{X,x}$ and similarly another triangle. This construction tells us that given any $k(x) \rightarrow K$, we have a morphism:

$$\text{Spec } K \rightarrow \text{Spec } k(x) \rightarrow \text{Spec } \mathcal{O}_{X,x} \rightarrow X$$

This gives us a bijection:

$$X(K) \simeq \bigsqcup_{x \in X} \text{Hom}_k(k(x), K)$$

Indeed, any $\text{Spec } K \rightarrow X$ factors through $\text{Spec } \mathcal{O}_{X,x} \rightarrow X$ uniquely. Then we have the following diagram:

$$\begin{array}{ccc}
\text{Spec } K & \xrightarrow{\quad} & X \\
& \searrow \exists! & \nearrow \\
& \text{Spec } \mathcal{O}_{X,x} &
\end{array}$$

Moreover, given any $\text{Spec } K \rightarrow \mathcal{O}_{X,x}$, we obtain a unique factorization:

$$\begin{array}{ccc}
\text{Spec } K & \xrightarrow{\quad} & \text{Spec } \mathcal{O}_{X,x} \\
& \searrow \exists! & \nearrow \\
& \text{Spec } k(x) &
\end{array}$$

In general, we will always consider an algebraic extension K/k , in this case, I claim that if $\text{Spec } K \rightarrow X$ is an k -morphism, then the image is a closed point. Indeed, suppose that x is the unique point in the image, we consider $\overline{\{x\}}$, which is given the canonical reduced closed subscheme structure. We consider $\dim \overline{\{x\}} = \text{trdeg}(k(x) : k) = 0$. Indeed, $\overline{\{x\}}$ is an integral finite type k -scheme, then the dimension is just the transcendence degree of the function field. Being dimension 0, it only has finitely many closed points, however since it's also irreducible, there is only one closed point. Therefore x is a closed point.

In particular, if K/k is of degree r , then:

$$X(K) = \bigsqcup_{\deg(x) \mid r} \text{Hom}_k(k(x), K)$$

Note that if $\deg(x) = e \mid r$, then there is a natural transitive action of $\text{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_q)$ on $\text{Hom}_k(k(x), K)$ which is defined by composition (Here we are thinking K and $k(x)$ are in the algebraic closure of $\mathbb{F}_q$).

For any $f \in \text{Hom}_k(k(x), K)$, the stabilize of f are those:

$$\sigma \circ f = f$$

Notice that $f : \mathbb{F}_{q^e} \rightarrow \mathbb{F}_{q^r}$ can be thought of as the restriction of some $F : \mathbb{F}_{q^r} \rightarrow \mathbb{F}_{q^r}$. Then we know that $\sigma \circ f = f$ iff:

$$\sigma \circ F|_{\mathbb{F}_{q^e}} = F|_{\mathbb{F}_{q^e}}$$

Notice that since $\mathbb{F}_{q^e}/\mathbb{F}_q$ is normal, the image of F is still in $\mathbb{F}_{q^e}$, therefore we can also think of f as an k -automorphism of $\mathbb{F}_{q^e}$. Then we can compose the above relation with the inverse to get:

$$\sigma \in \text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_{q^e})$$

Conversely, we know that any $\sigma \in \text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_{q^e})$ fixes f , therefore the stabilizer is just $\text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_{q^e})$. This also shows that the action is transitive since if g is another $\mathbb{F}_{q^e} \rightarrow \mathbb{F}_{q^r}$, then we can again think of it as some element in $\text{Gal}(\mathbb{F}_{q^e}|\mathbb{F}_q)$, then since each such element extends to some element in $\text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_q)$, then the action of $\text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_{q^e})$ is transitive. Then the orbit-stabilizer theorem tells us if $\deg(x) = e$:

$$|\text{Hom}_k(k(x), K)| = e$$

since $\text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_q)/\text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_{q^e}) = \text{Gal}(\mathbb{F}_{q^e}|\mathbb{F}_q)$.

Remark 1.2. If X is a variety over k and we have:

$$X = Y_1 \bigcup \cdots \bigcup Y_n$$

for locally closed subsets Y_i . Then if we give the canonical scheme structure on each Y_i , which is first give it's closure the reduced closed subscheme structure and use the fact that Y_i is open in the closure to give it the open subscheme structure. We know that the underlying set of these schemes are still Y_i and this construction doesn't affect the residue class field. Therefore we have:

$$X(K) = \bigcup_i Y_i(K)$$

If these Y_i are disjoint unions, then the above union should also be disjoint.

Remark 1.3. Given a variety over k and a finite extension K/k , since K is also a finite field, we know that $|X(K)|$ is in fact also finite.

Indeed, if X is affine, then we can assume that $X = \text{Spec} A$ where $A = k[a_1, \dots, a_n]$, then a point in $X(K)$ correspond to an k -algebra map:

$$k[a_1, \cdot, a_n] \rightarrow K$$

which are only finitely many. Then for a general X , cover it by finitely many affines and use the above remark.

Summarizing the above discussion, we have:

Proposition 1.4. If X is a variety over a finite field k and K/k is a finite extension of degree r , then $X(K)$ is finite and we have:

$$|X(K)| = \sum_{e|r} e \cdot |\{x \in X_{\text{cl}} : \deg(x) = e\}|$$

Proof. See the above discussion. □

It's often convenient to think of the base change of X to it's algebraic closure $\bar{k}$, which is $\bar{X} = X \times_k \bar{k}$. Indeed, this is still a variety over $\bar{k}$ and to show this we only need to show that reducedness is preserved. We make a short excursion to discuss this a bit as this is in general not proved in a first course of algebraic geometry. We follow the treatment on [Vak25].

1.3 Side Story on Reducedness

We begin with a theorem on transcendence degree:

Theorem 1.5. Any finitely generated extensions of perfect fields are **separably generated**. That is to say any finitely generated extension E/F of an perfect field F can be factored into a finite separable part and a purely transcendental part:

$$F \xrightarrow{\text{Transcendental}} F(t_1, \dots, t_n) \xrightarrow{\text{Finite Separable}} E$$

Any finitely generated field extension that can be factored in this way is said to be **separably generated**. This transcendence basis $\{t_1, \dots, t_n\}$ is said to be a **separating transcendence basis**.

*Proof. *** to be completed* □

Proposition 1.6. Suppose that B is a geometrically reduced k -algebra and A is a reduced k -algebra then $A \otimes_k B$ is reduced.

Proof. Let's first reduce the proof to the case where A is finitely generated over k . Indeed, to show $A \otimes_k B$ is reduced, we need to show whenever you have a nonzero $x \in A \otimes_k B$, $x^n \neq 0$ for all n . Since x is in the tensor product, we know that it's a finite linear combination of $a_i \otimes b_i$. Let A' be the subring generated over k by the elements a_i . Then $A' \otimes_k B \hookrightarrow A \otimes_k B$ is a subring. Then we know that $x^n = 0$ in $A \otimes_k B$ iff $x^n = 0$ in $A' \otimes_k B$.

In this case, we consider the canonical map:

$$A \rightarrow \prod_{\mathfrak{p}} A/\mathfrak{p}$$

where $\mathfrak{p}$ are finitely many minimal primes of A . Since A is reduced the kernel of this ring map is $\mathfrak{N}$, which is 0. Then we know that this map is injective. Then consider the injective map:

$$A \hookrightarrow \prod_{\mathfrak{p}} A/\mathfrak{p} \rightarrow \prod_{\mathfrak{p}} K(A/\mathfrak{p})$$

This shows that A is in fact a subring of the function fields of the irreducible components.

Then we know that if we can prove the following is reduced:

$$\left(\prod_{\mathfrak{p}} K(A/\mathfrak{p}) \right) \otimes_k B$$

Then $A \otimes_k B$ being a subring is also reduced. However, the product is reduced iff each $K(A/\mathfrak{p}) \otimes_k B$ is reduced. Therefore we can assume that A is just a field. But since B is geometrically reduced, we are done. □

Proposition 1.7. Suppose that A is a reduced k -algebra, then:

1. $A \otimes_k k(t)$ is reduced.
2. If E/k is a finite separable extension, then $A \otimes_k E$ is reduced.

Proof. To prove 1, note that $A \otimes_k k[t] = A[t]$ which is reduced. Then since $A \otimes_k k(t)$ is just a localization, we are done since localization preserves reducedness.

To prove 2, let's first assume E is generated by a single element with minimal polynomial $p(t)$ which is separable. We want to prove $E \otimes_k A$ is reduced, then we can again consider the subring A' of A finitely generated over k . Then again, we only need to assume that A is a field. Therefore we know that:

$$A \otimes_k k[t]/(p(t)) = A[t]/(p(t))$$

Since $p(t)$ is separable in $k[t]$, it's separable in $A[t]$, which implies that $A[t]/(p(t))$ is reduced.

Then for a general finite extension E/k , we can work inductively. □

Lemma 1.8. Suppose that E/k is a field extension of a perfect field k and A is a reduced k -algebra, then $A \otimes_k E$ is reduced.

Proof. First, by considering a subfield E' of E , we can assume that E is finitely generated over k . By the separating transcendence basis theorem, we know that E can be decomposed to something purely transcendental and something finite separable. Then we know that:

$$A \otimes_k k(t_1, \dots, t_n) \hookrightarrow A \otimes_k E = (A \otimes_k k(t_1, \dots, t_n)) \otimes_{k(t_1, \dots, t_n)} E$$

where $E/k(t_1, \dots, t_n)$ is finite separable. Notice that by the previous lemma:

$$A \otimes_k k(t_1, \dots, t_n)$$

is reduced, then $A \otimes_k E$ is reduced. □

Corollary 1.9. If E/k is a field extension of a perfect field, then X is reduced implies $X \otimes_k E$ is reduced.

Proof. Reducedness is a local property, then we can assume X is affine and apply the above lemma. $\square$

Corollary 1.10. Suppose that k is perfect and A, B are reduced k -algebras, then $A \otimes_k B$ is reduced.

Proof. Recall that A is geometrically reduced by the above discussion, then $A \otimes_k B$ is reduced. $\square$

Remark 1.11. Recall that given a field k and its algebraic closure $\bar{k}$, its perfect closure k^p is defined to be the smallest subextension k^p of $\bar{k}/k$ such that k^p is perfect and it's contained in all perfect extensions of k in $\bar{k}$.

Then what you proved above is that if X is a scheme over k , then if X_{k^p} is reduced, then X_K is reduced for all field extensions K of k . Again, we can assume that X is just affine and we want to show that:

$$A \otimes_k K$$

is reduced if A is reduced and $A \otimes_k k^p$ is reduced. Indeed, let's consider a perfect closure of K , say K^p , then we know that by the universal property of the perfect closure, there is a unique map of k -algebra from k^p to K^p . Then we know that:

$$A \otimes_k k^p \otimes_{k^p} K^p$$

is reduced by the above discussion. The above ring is just $A \otimes_k K^p$, which contains a subring $A \otimes_k K$, thus $A \otimes_k K$ is also reduced.

Therefore we know that the following statements are equivalent:

1. X_K is reduced for all field extensions K/k .
2. X_K is reduced for all field extensions K/k such that $K = \bar{K}$.
3. $X_{\bar{k}}$ is reduced.
4. X_{k^p} is reduced.

We also have equivalent statements for irreducibility and connectedness, but we won't include it here.

In conclusion, we proved that schemes over perfect fields are geometrically reduced.

This ends our short excursion to reducedness.

1.4 Back to Varieties Over Finite Fields

Again we let $k = \mathbb{F}_q$ and now we consider an algebraic closure $\bar{k}$, since k is a finite field, then it's already perfect, therefore we know that $\bar{X} = X \otimes_k \bar{k}$ is again reduced by the previous discussion, thus a variety.

Note that by definition:

$$\bar{X}(\bar{k}) = \text{Hom}_{\bar{k}}(\text{Spec } \bar{k}, \bar{X})$$

By the universal property of fibered product, this is identified with $\text{Hom}_k(\text{Spec } \bar{k}, X)$.

Besides the arithmetic and geometric Frobenius on $\text{Gal}(\bar{k}/k)$, we have an **absolute Frobenius morphism** on X .

$$\text{Frob}_{X,q} : X \rightarrow X$$

This is defined by the identity map on X as a map of topological spaces and on the sheaves of rings it's just:

$$\mathcal{O}_X \rightarrow \mathcal{O}_X, \quad u \mapsto u^q$$

This is well defined again because X is over k , and this definition gives a morphism of k -schemes since u^p fixes $u \in k$.

Note that it induces a morphism of $\bar{k}$ -schemes on $\bar{X}$ defined by:

$$\text{Frob}_{\bar{X},q} : \bar{X} \rightarrow \bar{X}$$

This morphism is usually written as $\text{Frob}_{X,q} \times_k \text{Id}_{\bar{k}}$. Indeed, we know that by the universal property of fibered products, give a morphism from X to X and a morphism from $\text{Spec} \bar{k}$ to $\text{Spec} \bar{k}$ such that they agree after goes to $\text{Spec} k$ gives a morphism $\bar{X} \rightarrow \bar{X}$, which can be heuristically thought of as give what the morphism does on two coordinates. This is illustrated in the following diagram:

$$\begin{array}{ccccc} \bar{X} & \longrightarrow & \text{Spec} \bar{k} & & \\ \downarrow & \searrow & \downarrow \text{Id} & \searrow & \\ X & & \bar{X} & \longrightarrow & \text{Spec} \bar{k} \\ & \searrow \text{Frob}_{X,q} & \downarrow & & \downarrow \\ & & X & \longrightarrow & \text{Spec} k \end{array}$$

This is a functorial construction. Indeed, we can pullback any k -variety morphism f to $\bar{f}$. And if we have a morphism of k -varieties:

$$f : X \rightarrow Y$$

one gets the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{\text{Frob}_{X,q}} & X \\ \downarrow f & & \downarrow f \\ Y & \xrightarrow{\text{Frob}_{Y,q}} & Y \end{array}$$

I will let you check it indeed commutes (This in the language of category theory, says that $\text{Frob}_{X,q}$ is a natural transformation between the identity functors of k -varieties). Then we know that after we pull this back along $\bar{k}$, we get:

$$\begin{array}{ccc} \bar{X} & \xrightarrow{\text{Frob}_{\bar{X},q}} & \bar{X} \\ \downarrow \bar{f} & & \downarrow \bar{f} \\ \bar{Y} & \xrightarrow{\text{Frob}_{\bar{Y},q}} & \bar{Y} \end{array}$$

Using the universal property of fibered products and think of the pullback as give two morphisms on two factors, you can easily see this commutes.

In particular, if we have some closed embedding $X \hookrightarrow \mathbb{A}_k^n$, then we know that after we pullback, the Frobenius morphism on $\bar{X}$ is induced by the Frobenius on $\mathbb{A}_k^n$.

Example 1.12. Let $X = \mathbb{A}_k^n$, let's try to understand what is $\text{Frob}_{X,q}$ on X .

Notice that it correspond to the ring morphism:

$$k[x_1, \dots, x_n] \rightarrow k[x_1, \dots, x_n], \quad f(x) \mapsto (f(x))^q$$

Now if we pull it back to $\bar{k}$, we get a morphism:

$$\text{Spec} \bar{k}[x_1, \dots, x_n] \rightarrow \text{Spec} \bar{k}[x_1, \dots, x_n]$$

Notice that by definition, $\bar{k}$ is mapped to $\bar{k}$ via identity, but x_i is mapped to x_i^q .

Note that suppose that each morphism of k -schemes $f : X \rightarrow Y$ gives us a map of K -points by composition:

$$X(k) \rightarrow Y(K)$$

Then if we think about what $\text{Frob}_{\overline{X},q}$ does on the $\overline{k}$ -points it just correspond to the point $(u_1, \dots, u_n)$ is mapped to $(u_1^q, \dots, u_n^q)$.

Recall that $\overline{X}(\overline{k}) = X(\overline{k})$. Then we know that when $X = \mathbb{A}_k^n$, if we think about the embedding:

$$X(\mathbb{F}_{q^r}) \hookrightarrow X(\overline{k}) = \overline{X}(\overline{k})$$

We know that the image are exactly those points in $\overline{X}(\overline{k})$ fixed by $\text{Frob}_{\overline{X},q}^r$.

Similarly, if X is affine, we obtain the same conclusion by think of X as a closed variety of $\mathbb{A}_k^n$. Then we have:

$$\begin{array}{ccc} X(\mathbb{F}_{q^r}) & \hookrightarrow & X(\overline{k}) \\ \downarrow & & \downarrow \\ \mathbb{A}_k^n(\mathbb{F}_{q^r}) & \hookrightarrow & \mathbb{A}_k^n(\overline{k}) \end{array}$$

This implies that $X(\mathbb{F}_{q^r})$ is indeed fixed by $\text{Frob}_{\overline{X},q}^r$, conversely, if a point in $X(\overline{k})$ is fixed by all $\text{Frob}_{\overline{X},q}^r$, then clearly it's in $X(\mathbb{F}_{q^r})$.

Now when X is not affine, we can take affine cover U_i of X and we know that:

$$X(\overline{k}) = \bigcup_i U_i(\overline{k}), \quad X(\mathbb{F}_{q^r}) = \bigcup_i U_i(\mathbb{F}_{q^r})$$

We know that for any $x \in X(\overline{k})$ it's in some $U_i(\overline{k})$. First assume that it's fixed by $\text{Frob}_{\overline{X},q}^r$. One interesting property of the Frobenius $\text{Frob}_{X,q}$ on X is that if you restrict it to U where U is any open subscheme of X , we get $\text{Frob}_{U,q}$. Similarly, we know that if we restrict $\text{Frob}_{\overline{X},q}$ to $\overline{U}$, we get $\text{Frob}_{\overline{U},q}$. Therefore if x is fixed by $\text{Frob}_{\overline{X},q}^r$ and $x \in U_i(\overline{k})$, we know that it also is fixed by $\text{Frob}_{U_i,q}$, therefore it's in $U_i(\mathbb{F}_{q^r})$, thus in $X(\mathbb{F}_{q^r})$. Conversely, we know that if $x \in X(\mathbb{F}_{q^r})$, then it's in $U_i(\mathbb{F}_{q^r})$ for some U_i and it's fixed by $\text{Frob}_{U_i,q}^r$, which means that it's fixed by $\text{Frob}_{\overline{X},q}^r$ since we have:

$$\begin{array}{ccccc} \text{Spec } \overline{k} & \longrightarrow & \overline{U}_i & \xrightarrow{\text{Frob}_{\overline{U}_i,q}^r} & \overline{U}_i \\ & & \downarrow & & \downarrow \\ & & \overline{X} & \xrightarrow{\text{Frob}_{\overline{X},q}^r} & \overline{X} \end{array}$$

I will let you do the details but essentially it's just the fact open embeddings are monomorphisms.

In conclusion, we proved that:

Proposition 1.13. Let X be a variety over k , then we have:

$$X(\mathbb{F}_{q^r}) = (X(\overline{k}))^{\text{Frob}_{\overline{X},q}^r}$$

Proof. See the above discussion. □

This interpretation of $X(\mathbb{F}_{q^r})$ is already very useful since arithmetic point of view because we can use the Frobenius. But we have another geometric interpretation which connects to the classic intersection theory:

Proposition 1.14. Let Δ, Γ_r be the diagonal of $\overline{X} \times_k \overline{X}$ and graph of $\text{Frob}_{\overline{X},q}^r : \overline{X} \rightarrow \overline{X}$, then we have a natural bijection between the closed ($\overline{k}$ -points) points of the scheme theoretic intersection $\Delta \cap \Gamma_r$.

In other words, $\overline{k}$ -points of $\Delta \cap \Gamma_r$ are in bijection with $X(\mathbb{F}_{q^r})$.

Proof. The intersection is given by the following Cartesian diagram:

$$\begin{array}{ccc} \Delta \cap \Gamma_r & \hookrightarrow & \Gamma_r \\ \downarrow & & \downarrow \\ \Delta & \hookrightarrow & \overline{X} \times_{\overline{k}} \overline{X} \end{array}$$

Recall that the diagonal and the graph are all locally closed embeddings. It's easy to see that the diagonal is a locally closed embedding and to see that the graph is a locally closed embedding, and you can also work locally to show that the graph is also a locally closed embedding for all morphisms. Indeed consider the diagram:

$$\begin{array}{ccccc}
 X & & & & \\
 \swarrow \Gamma_f & \searrow f & & & \\
 & X \times_S Y & \longrightarrow & Y & \\
 \downarrow \text{Id} & \downarrow & & \downarrow & \\
 & X & \longrightarrow & S &
 \end{array}$$

We know that if we cover Y by affine opens $\text{Spec} B$ and cover the inverse image under $f : X \rightarrow Y$ by $\text{Spec} A$, one checks $\text{Spec} A \otimes_C B$ covers the image of Γ_f . On each of these affine open, the graph looks like:

$$\text{Spec} A \rightarrow \text{Spec} A \otimes_C B$$

which is indeed surjective.

Therefore the intersection of $\Delta \cap \Gamma_f$ defined by Cartesian product makes sense. We know that since Δ and Γ_f are both locally finite type over $\bar{k}$, therefore we know that the $\bar{k}$ -points are exactly the closed points (see the remark afterwards if you don't feel confident about this statement). Therefore in the statement, we can use $\bar{k}$ -valued points instead of closed points.

Then we know that if we consider a $\bar{k}$ -scheme map from $\text{Spec} \bar{k}$ to $\Delta \cap \Gamma_f$, it's defined by two $\bar{k}$ -morphisms a, b such that $\Delta \circ a = \Gamma_f \circ b$:

$$\{(a, b) \in \text{Hom}_{\bar{k}}(\text{Spec} \bar{k}, \bar{X}) \times \text{Hom}_{\bar{k}}(\text{Spec} \bar{k}, \bar{X}) : \Delta \circ a = \Gamma_f \circ b\}$$

Note that since $\Delta \circ a$ is defined by the following diagram:

$$\begin{array}{ccccc}
 \text{Spec} \bar{k} & \xrightarrow{a} & \bar{X} & & \\
 & \searrow \Delta & \searrow \text{Id} & & \\
 & & \bar{X} \times_{\bar{k}} \bar{X} & \longrightarrow & \bar{X} \\
 & \searrow \text{Id} & \downarrow & & \downarrow \\
 & & \bar{X} & \longrightarrow & \text{Spec} \bar{k}
 \end{array}$$

Therefore such $\Delta \circ a$ can be thought of as defined by a single $\bar{k}$ -morphism $a : \text{Spec} \bar{k} \rightarrow \bar{X}$. Similarly, we know that $\Gamma_f \circ b$ is defined by:

$$\begin{array}{ccccc}
 \text{Spec} \bar{k} & \xrightarrow{b} & \bar{X} & & \\
 & \searrow \Gamma_r & \searrow \text{Frob}_{\bar{X}, q}^r & & \\
 & & \bar{X} \times_{\bar{k}} \bar{X} & \longrightarrow & \bar{X} \\
 & \searrow \text{Id} & \downarrow & & \downarrow \\
 & & \bar{X} & \longrightarrow & \text{Spec} \bar{k}
 \end{array}$$

which is a single $\bar{k}$ -morphisms $b : \text{Spec} \bar{k} \rightarrow \bar{X}$ such that $b = \text{Frob}_{\bar{X}, q}^r \circ a$. Therefore we know that a, b must satisfy that:

$$(a, a) = (b, \text{Frob}_{\bar{X}, q}^r \circ b)$$

This implies $a = b$ and $a = \text{Frob}_{\bar{X}, q}^r \circ a$, which means that a should be a point in $\bar{X}(\bar{k})$ such that it's fixed by the $\text{Frob}_{\bar{X}, q}^r$ action. This defines a bijection due to the universal properties. $\square$

Remark 1.15. *** to be completed

The key here is that if x is a closed point, then the structure morphism $\bar{k} \rightarrow k(x)$ is itself an isomorphism!

The following result makes us understand the intersection $\Delta \cap \Gamma_r$ better:

Proposition 1.16. If X is smooth over k , then $\Delta \cap \Gamma_r$ intersect transversely, or in other words, the scheme theoretic intersection $\Delta \cap \Gamma_r$ consist of only a finite set of reduced points.

Remark 1.17. Since k is a perfect field, then smoothness is equivalent to being nonsingular, we will include a proof after we finish this result.

Proof. Let's consider a simple case where $X = \mathbb{A}_{\mathbb{F}_q}^n$. In this case if we let $R = \bar{k}[x_1, \dots, x_n, y_1, \dots, y_n]$. I will let you show that Δ in R is defined by $(x_1 - y_1, \dots, x_n - y_n)$. The graph Γ_r is defined by $(y_1 - x_1^{q^r}, \dots, y_n - x_n^{q^r})$. In this case $\Gamma_r \cap \Delta$ is isomorphic to:

$$k[x_1, \dots, x_n] / (x_1 - x_1^{q^r}, \dots, x_n - x_n^{q^r})$$

Now notice that this is isomorphic to a tensor product:

$$\bigotimes_{\bar{k}}^i \bar{k}[x_i] / (x_i - x_i^{q^r})$$

Notice that each factor $x_i - x_i^{q^r}$ is separable and splits into linear factors, therefore we know that by the Chinese Remainder Theorem this is just a product of $\bar{k}$, which is a finite reduced set of points.

To show the general case, we will need the following lemma:

Lemma 1.18. Suppose that X is an integral finite type $\bar{k}$ -scheme over $\bar{k}$ with positive dimension, then it has infinitely many closed points.

Proof. Notice that it suffices to prove this for affine integral $\bar{k}$ varieties. Suppose that $X = \text{Spec} A$. Recall that the dimension of X is $\text{trdeg} K(X) > 0$, note that by clearing out denominators, we can assume that one of the transcendence basis x is actually in A , then we know that x and some other elements generate A as an k -algebra. Then consider the map of $\bar{k}$ -algebra:

$$A \rightarrow \bar{k}$$

defined by mapping x to a and other generators to 0 is well defined. But since $\bar{k}$ is infinite, we know that there are infinitely many such closed points. $\square$

This lemma tells you that only a zero dimensional $\bar{k}$ variety has finitely many closed points. Then for an arbitrary $\bar{k}$ -variety X , we already know that $\Delta \cap \Gamma_r$ has only finitely many closed points, this shows that the intersection is already 0-dimensional. We only need to show that each point is reduced.

Let $u \in X(\mathbb{F}_{q^r})$ and let x be the corresponding closed point. Recall that we assumed that X is smooth and this is equivalent of X being nonsingular, which means that x is a regular point. Then there will be a system parameters $t_1, \dots, t_n \in \mathcal{O}_{X,x}$ and we know that these parameters defines an étale morphism:

$$U \rightarrow \mathbb{A}_k^n$$

where U is an open neighborhood of x . Recall that each such $\mathbb{F}_{q^r}$ -valued point correspond to a closed point in $\Delta \cap \Gamma_r$ which lies in $\bar{U} \times \{U\}$. Indeed, we have the following diagram:

$$\begin{array}{ccccc} & \text{Spec } \bar{k} & & & \\ & \downarrow & \searrow & \searrow & \\ \bar{U} & \hookrightarrow & \bar{X} & \longrightarrow & \text{Spec } \bar{k} \\ & \downarrow & \downarrow & & \downarrow \\ U & \hookrightarrow & X & \longrightarrow & \text{Spec } k \end{array}$$

(Note: A dashed curved arrow also points from $\text{Spec } \bar{k}$ down to U .)

This shows that a $\bar{X}(\bar{k})$ point, no matter it's fixed by $\text{Frob}_{\bar{X},q}^r$ or not, gives the $X(\bar{k})$, then if it's fixed by $\text{Frob}_{X,q}^r$, then the point in $X(\bar{k})$ is actually in $X(\mathbb{F}_{q^r})$.

Therefore if we can show that $\Delta \cap \Gamma_r$ in $\bar{U} \times \{U\}$ contains a finite number of reduced points we are done. Indeed, this is given in the following lemma:

Lemma 1.19. Let X be a nonsingular variety over k where x is a closed point such that $k(x)/k$ is separable. Then if we take $t_1, \dots, t_n$ to be a regular system of parameters of A , then the germs can be represented by some regular functions on an affine open neighborhood of x in X . They also define an étale morphism:

$$\varphi : U \rightarrow \mathbb{A}_k^n, \quad \varphi(x) = 0$$

Proof. Indeed, we can assume that each of the parameter comes from some affine open near x , which basically is the definition of the germ. Then we can pick an affine open in the intersection such that these parameters are indeed regular functions. If n is 0, which means that X itself near x (precisely speaking it's the irreducible component containing x) is just a 0-dimensional. In this case, we know that x is just a discrete point and we can just choose $U \rightarrow \mathbb{A}_k^n$ to be the morphism induced by $k \rightarrow k(x)$.

Therefore, let's assume that the regular parameter is not empty and they live on $\text{Spec} B$, then we can consider a map of k -algebras:

$$k[x_1, \dots, x_n] \rightarrow B, \quad x_i \mapsto t_i$$

I claim that this map induces an étale morphism $\text{Spec} B \rightarrow \mathbb{A}_k^n$.

It's first easy to see that using this definition, x is mapped to the origin in $\mathbb{A}_k^n$. Indeed, if we consider the inverse image of x , since we can assume that t_i generates the maximal ideal corresponding to x in $\text{Spec} B$, then we know that the inverse image contains $(t_1, \dots, t_n)$, which is itself a maximal ideal.

Then we need to show this morphism is indeed étale. *** review differentials and then come back later $\square$

Using the above result, we consider the following étale morphism since base change of étale morphisms are still étale:

$$\overline{U} \times \overline{U} \rightarrow \mathbb{A}_k^n \times \mathbb{A}_k^n$$

Notice that we can consider the intersection of the diagonal and the graph in $\overline{U} \times \overline{U}$, which is just the restriction of the intersection in $\mathbb{A}_k^n \times \mathbb{A}_k^n$ which means that the following is Cartesian:

$$\begin{array}{ccc} \Delta \cap \Gamma_r & \longrightarrow & \Delta \cap \Gamma_r \\ \downarrow & & \downarrow \\ \overline{U} \times \overline{U} & \longrightarrow & \mathbb{A}_k^n \times \mathbb{A}_k^n \end{array}$$

Notice that we know that $\Delta \cap \Gamma_r$ is a finite set of reduced points in the affine space and the morphism below is étale. This implies that the map above is étale which means smooth of relative dimension 0, but we know that $\Delta \cup \Gamma_r$ is itself smooth (reduced) of relative dimension 0 over $\overline{k}$, therefore so is the product. *** need to fill in some details $\square$

1.5 Nonsingular and Smooth Varieties over Perfect Fields

The main goal of this yet another short excursion is to prove the following important result:

Theorem 1.20. Let X is a finite type k -scheme:

1. If k is perfect then every regular finite type k -scheme is smooth over k .
2. Every smooth k -scheme is regular with no perfect field assumption.

Technically speaking, since 2 implies 1, for the purpose of the above proposition of transverse intersection, we only need 2, thus we only prove $2 \rightarrow 1$ which is the harder direction as it uses more machinery of étale morphisms:

Proposition 1.21. Let X is a finite type k -scheme, then Every smooth k -scheme is regular with no perfect field assumption.

We will need the following result, which can be very useful when we try to determine regularity:

Proposition 1.22. Suppose that X is a Noetherian scheme and D is an effective Cartier divisor of X and $p \in D$. If p is regular in D , then p is regular in X .

Proof. This whole proposition is local, therefore we can assume $X = \text{Spec} A$ where A is a Noetherian ring and $D = \text{Spec} A/t$ where t is a nonzero divisor in A . Let $\mathfrak{p}$ be a prime over (t) , but our assumption $\mathfrak{p}/(t)$ is regular. In fact, we can assume that A is a Noetherian local ring and $\mathfrak{p}$ a the unique maximal ideal. Indeed, we know that since $t \in \mathfrak{p}$, we know that after we localize at $\mathfrak{p}$, t is still a nonzero divisor since if $t \cdot \frac{a}{b} = 0$, which implies that $tay = 0$ where $y \notin \mathfrak{p}$, then we know that since t is not a zero divisor, $ay = 0$, but this implies that $\frac{a}{b} = 0$.

Let's assume that $A/(t)$ is regular of dimension n (Since it's Noetherian and we know that Noetherian local rings always have finite dimension). Therefore our assumption now becomes:

$$\mathfrak{p}/(t) + \mathfrak{p}^2$$

is a dimension n $A/\mathfrak{p}$ vector space. Notice that since t is not a zero divisor, we know that by Krull's theorem A is of dimension $n + 1$. We want to show that $\mathfrak{p}/\mathfrak{p}^2$ is a dimension $n + 1$ vector space. Notice that if we pick out $x_1, \dots, x_n$ which is a basis of $\mathfrak{p}/(t) + \mathfrak{p}^2$ over $A/\mathfrak{p}$, we notice that $t, x_1, \dots, x_n$ will span $\mathfrak{p}/\mathfrak{p}^2$, then we only need to show they are indeed linearly independent. If:

$$k_1 x_1 + \dots + k_n x_n + kt = 0 \in \mathfrak{p}/\mathfrak{p}^2$$

We can further quotient out by t and we get that $k_1, \dots, k_n = 0$. In this case it implies that $t \in \mathfrak{p}^2$. However if $t \in \mathfrak{p}^2$, $A/(t)$ can not be regular since we know that for every Noetherian local ring, the existence of local parameter forces the dimension of the ring to be no larger than the dimension of the tangent space. Therefore we know that:

$$n + 1 = \dim A \leq \dim \mathfrak{p}/\mathfrak{p}^2 = \dim \mathfrak{p}/(t) + \mathfrak{p}^2 = n$$

Therefore $\mathfrak{p}$ is regular at A . □

Remark 1.23. Notice that the above statement can be completely algebraic and we can just say that $(A, \mathfrak{m})$ is a Noetherian local ring where t is a nonzerodivisor such that A/t is regular of dimension n , then A is regular of dimension $n + 1$. Noetherian hypothesis is really important since it makes sure the dimension of the ring is always at most the dimension of the tangent space.

Note that if $(A, \mathfrak{m})$ is a regular local ring of dimension n (Recall that when we say a regular local ring, Noetherian are by default in the definition) and $t \in A$ is a nonzero divisor, then A/t is of dimension $n - 1$ using Krull's Theorem since A is Noetherian, then clearly the tangent space of A/t is 1 less, therefore A/t is also a regular local ring of dimension $n - 1$.

This combines with the above proposition, we get a two way implication.

Using this (and of course assuming a lot of properties of étale morphisms which we can not prove here) we can prove the following:

Proposition 1.24. Let $\pi : X \rightarrow Y$ be an étale morphism. Suppose that $q \in Y$ is a regular point whose local ring has dimension n , then any point p in $\pi^{-1}q$ is also a regular point with local ring of dimension n .

Proof. By our assumption, we know that the map of local rings:

$$\mathcal{O}_{Y,q} \rightarrow \mathcal{O}_{X,p}$$

is flat thus automatically faithfully flat.

Recall that since π is étale, the fiber of q is just a disjoint union of points, each of which a finite separable extension of $\kappa(q)$. This implies that if you think about what is $\mathcal{O}_{X,p}$, it's actually a field. Indeed, if we assume that locally the morphism is:

$$\text{Spec} B \rightarrow \text{Spec} A$$

we know that the fiber is:

$$\text{Spec} B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}} \rightarrow \text{Spec} A_{\mathfrak{q}}/\mathfrak{q}A_{\mathfrak{q}}$$

Here $\text{Spec} B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}}$ is just a disjoint union of finite separable extensions of $A_{\mathfrak{q}}/\mathfrak{q}A_{\mathfrak{q}}$. Clearly if $\mathfrak{p}$ is a prime of B which maps to $\mathfrak{q}$. Then we know that each of these $\mathfrak{p}$ correspond to a maximal ideal in:

$$\prod_i K_i$$

where K_i is a finite separable extension of $A_{\mathfrak{q}}/\mathfrak{q}A_{\mathfrak{q}}$. Localize at $\mathfrak{p}$, we get:

$$B_{\mathfrak{p}}/\mathfrak{q}B_{\mathfrak{p}}$$

We know this is a field.

Then let's prove that $\mathcal{O}_{X,p}$ is regular by induction. First if $\mathcal{O}_{Y,p}$ is regular of dimension 0, then it's a field since $\mathfrak{m}_q = 0$. In this case we know that:

$$B_{\mathfrak{p}}/\mathfrak{q}B_{\mathfrak{p}} = B_{\mathfrak{p}}$$

since $\mathfrak{q} = 0$, which is a field, therefore regular of dimension 0. Now notice that since this is a field, then we know that $\mathfrak{q}B_{\mathfrak{p}} = \mathfrak{p}B_{\mathfrak{p}}$, and this implies that $\mathfrak{m}_p = \mathcal{O}_{X,p}\mathfrak{m}_q$.

Now assumed that we have proved this case for $\mathcal{O}_{Y,q}$ with dimension at most $n-1$. If now it's dimension n , let's pick an element t in $\mathfrak{m}_q/\mathfrak{m}_q$, we know since each regular local ring is an integral domain, t is not a zero divisor. Then consider it's image in $\mathcal{O}_{X,p}$. Consider the ring map:

$$\mathcal{O}_{Y,q}/t \rightarrow \mathcal{O}_{X,p}/t$$

From the above remark, we know that $\mathcal{O}_{Y,q}/t$ is regular local of dimension $n-1$. Notice that since $\mathcal{O}_{Y,q} \rightarrow \mathcal{O}_{X,p}$ is flat, then t is not a zero divisor in $\mathcal{O}_{X,p}$. Notice we can assume that t is a regular function lives on some open neighborhood, then we know that by shrinking the domain and codomain of π , we can think that π is :

$$\pi : \text{Spec} B \rightarrow \text{Spec} A$$

where $t \in A$, then we know that we have the pullback of effective divisor $\text{Spec} A/t \rightarrow \text{Spec} A$, which is $\text{Spec} B/t \rightarrow \text{Spec} A/t$ which is still étale since pullback of étale morphisms are still étale. Then by hypothesis, we know that $\mathcal{O}_{X,p}/t$ is going to be regular local of dimension n and t is not a zero divisor, then the above remark tells you that $\mathcal{O}_{X,p}$ is regular local of dimension n . We are done. $\square$

Finally let's prove 1.21:

Proof. Suppose that X is a k -scheme that is smooth over k . By definition of smoothness, we know that locally this morphism can be written as:

$$\text{Spec} k[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r) \rightarrow \text{Spec} k$$

such that the Jacobian of f_i with respect to the first r x_i is invertible on $\text{Spec} k[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r)$. Then if we consider the following diagram:

$$\begin{array}{ccc} \text{Spec} k[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r) & \xrightarrow{\quad\quad\quad} & \text{Spec} k \\ & \searrow & \nearrow \\ & \text{Spec} k[x_{r+1}, \dots, x_{r+n}] & \end{array}$$

Notice that the morphism:

$$\text{Spec} k[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r) \rightarrow \text{Spec} k[x_{r+1}, \dots, x_{r+n}]$$

is étale. Indeed, this is basically just by definition. Then recall that $\text{Spec} k[x_{r+1}, \dots, x_{r+n}]$ will be regular, then we can use the above result from étale morphism and conclude that $\text{Spec} k[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r)$ is regular. $\square$

1.6 Some Exercises

In this section, let X and $\bar{X}$ as our usual convention, and $G = \text{Gal}(\bar{k}/k)$ acts on the right of $\text{Spec} \bar{k}$ as $\text{Spec} k$ morphisms, which is basically that the action of $\sigma \in G$ is the automorphism it induces on $\text{Spec} \bar{k}$. Then we know that if you first apply the action of σ_1 then σ_2 , it's actually the automorphism induced by $\sigma_1 \circ \sigma_2$, therefore the natural action should be a right action.

Notice that for each σ , we can consider the morphism:

$$\text{Id}_X \times_k \sigma : \overline{X} \rightarrow \overline{X}$$

Notice that this is indeed an automorphism of $\overline{X}$ over k and clearly G also acts on the right $\overline{X}$ via this induced automorphism.

Problem 1.25. Try to describe the action of the arithmetic Frobenius τ on $\overline{X}$ when $X = \mathbb{A}_k^n$.

Solution. By definition, we know that the action is described by the following diagram:

$$\begin{array}{ccccc} \overline{k}[x_1, \dots, x_n] & \longleftarrow & \overline{k} & \xleftarrow{x \mapsto x^q} & \overline{k} \\ & \nwarrow & & \nearrow & \\ k[x_1, \dots, x_n] & & \overline{k}[x_1, \dots, x_n] & \longleftarrow & \overline{k} \\ & \nwarrow \text{Id} & \nearrow & & \\ & & k[x_1, \dots, x_n] & \longleftarrow & k \end{array}$$

Then you find that this is actually described by the following k -map:

$$\overline{k}[x_1, \dots, x_n] \rightarrow \overline{k}[x_1, \dots, x_n]$$

where $x \mapsto x^q$ if $x \in \overline{k}$ and $x_i \mapsto x_i$. You can check that this is indeed a ring map and fixes k .

Problem 1.26. Now assume that X is any variety over k , then show:

$$\tau \circ \text{Frob}_{\overline{X}, q} = \text{Frob}_{\overline{X}, q} \circ \tau$$

which both equal to the absolute Frobenius on $\overline{X}$, where τ is again the action induced by the arithmetic Frobenius.

Solution. Notice that if X is $\mathbb{A}_k^n$ then recall that the $\text{Frob}_{\overline{X}, q}$ is $\overline{k}[x_1, \dots, x_n] \rightarrow \overline{k}[x_1, \dots, x_n]$ which fixes $\overline{k}$ but takes x_i to x_i^q . The action τ is $\overline{k}[x_1, \dots, x_n] \rightarrow \overline{k}[x_1, \dots, x_n]$ which fixes $\overline{k}$ but this time fixes x_i and maps $x \in \overline{k}$ to x^q , then clearly the composition is just:

$$\overline{k}[x_1, \dots, x_n] \rightarrow \overline{k}[x_1, \dots, x_n], \quad f \mapsto f^q$$

Similarly with the other direction.

Now if $X \hookrightarrow \mathbb{A}_k^n$ is a closed embedding, we know that via similar argument, you find that the action of τ on $\overline{X}$ is again fix the variables but maps elements in $\overline{k}$ to q -th power.

Now notice that the above construction of the action by τ is functorial in the sense that if we have some morphism of k -varieties $f : X \rightarrow Y$, then we can consider the following diagram:

$$\begin{array}{ccc} \overline{X} & \xrightarrow{f \times \text{Id}} & \overline{Y} \\ \tau \downarrow & & \downarrow \tau \\ \overline{X} & \xrightarrow{f \times \text{Id}} & \overline{Y} \end{array}$$

Again this is because the fibered product is a functorial construction.

In particular, if U is an open subscheme of X , then we know that we have:

$$\begin{array}{ccc} \overline{U} & \hookrightarrow & \overline{X} \\ \tau \downarrow & & \downarrow \tau \\ \overline{U} & \hookrightarrow & \overline{X} \end{array} \quad \text{This tells one}$$

that the restriction of τ on $\overline{X}$ to $\overline{U}$ is the action of τ directly defined on $\overline{U}$.

Recall that we also have the similar results for $\text{Frob}_{\overline{U}, q}, \text{Frob}_{\overline{X}, q}$. Therefore we know that to check the assertion on $\overline{X}$, it suffices to check it on $\overline{U}$, where U is an affine open, which is the affine case we did above.

Notice that we also have a natural left action of G on $X(\bar{k})$ which is defined by:

$$(g, \phi) = \phi \circ g$$

You check that this is indeed a left action since:

$$(g_1 g_2, \phi) = (\phi \circ g_2) \circ g_1 = (g_1, (g_2, \phi))$$

Problem 1.27. Show the action of the arithmetic Frobenius on $X(\bar{k}) = X(\bar{k})$ is the same as the action induced by $\text{Frob}_{\bar{X}, q}$.

Solution. Recall that if $\phi : \text{Spec } \bar{k} \rightarrow \bar{X}$ is a $\bar{k}$ -point of $\bar{X}$, then the action of $\text{Frob}_{\bar{X}, q}$ is just:

$$(\text{Frob}_{\bar{X}, q}, \phi) = \text{Frob}_{\bar{X}, q} \circ \phi$$

In other words we are trying to show that:

$$\text{Frob}_{\bar{X}, q} \circ \phi = \phi \circ \tau$$

Clearly, it suffice to assume X is affine since we can to prove two maps from $\text{Spec } \bar{k}$ to $\bar{X}$ are the same, it suffices to prove it if we choose an affine open cover of $\bar{X}$. Notice that then we can assume that ϕ is defined by:

$$\bar{k}[x_1, \dots, x_n]/(f_1, \dots, f_r) \rightarrow \bar{k}, \quad x_i \mapsto k_i$$

Then we know that if we post compose it with $\text{Frob}_{\bar{X}, q}$, we just get:

$$\bar{k}[x_1, \dots, x_n]/(f_1, \dots, f_r) \rightarrow \bar{k}[x_1, \dots, x_n]/(f_1, \dots, f_r) \rightarrow \bar{k}$$

where the first map is defined by $x_i \mapsto x_i^q$. (Now you see it is very important to require $\bar{k}$ an algebraic closed field since otherwise this map may not be well defined). The composition is just $x_i \mapsto k_i^q$.

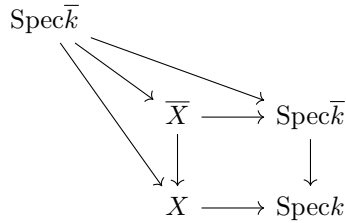
Now we consider $\phi \circ \tau$, which is induced by the map:

$$\bar{k}[x_1, \dots, x_n]/(f_1, \dots, f_r) \rightarrow \bar{k} \rightarrow \bar{k}$$

The first map is still ϕ however, the second map will be $k_i \mapsto k_i^q$, therefore we know that this maps is again defined by $x_i \mapsto k_i^q$.

Note that the canonical projection $\bar{X} \rightarrow X$ actually induces a map $\bar{X}_{\text{cl}} \rightarrow X_{\text{cl}}$. Indeed, we know that if $\bar{x}$ is mapped to x , we get an k -extension $k(x) \rightarrow \bar{k}$, which means that $k(x)$ is finitely generated and algebraic, thus finite, which further implies x is also closed.

Problem 1.28. If we identify the closed points of $\bar{X}$ as the $\bar{k}$ -valued points of $\bar{X}$, then the above map $\bar{X}_{\text{cl}} \rightarrow X_{\text{cl}}$ is identified with the map that takes $a \in X(\bar{k})$ to the image of a .



Solution. This is basically the below diagram:

Now if we think about the group $\text{Gal}(\mathbb{F}_{q^r}/k)$, notice that it acts on the left on $X(\mathbb{F}_{q^r})$. We also have a map $X(\mathbb{F}_{q^r}) \rightarrow X_{\text{cl}}$ which is defined by taking $a \in X(\mathbb{F}_{q^r})$ and map it to its image.

Problem 1.29. The fibers of the map $X(\mathbb{F}_{q^r}) \rightarrow X_{\text{cl}}$ are the orbits of the $\text{Gal}(\mathbb{F}_{q^r}/k)$ action.

Solution. If $k(x)/k$ is of degree e but $e \nmid r$, then clearly there is no $a \in X(\mathbb{F}_{q^r})$ such that it's image is x . Therefore let's assume that there is at least one $a \in X(\mathbb{F}_{q^r})$ such that x is the image of a . Then recall that by our previous discussion, all the $X(\mathbb{F}_{q^r})$ with image x , which is the fiber of the map in the statement can be described by:

$$\text{Hom}_k(k(x), \mathbb{F}_{q^r})$$

Let's consider the orbit of a , we are just then trying to show that the action of $\text{Gal}(\mathbb{F}_{q^r}/k)$ on $\text{Hom}_k(k(x), \mathbb{F}_{q^r})$ is transitive, but we already know this with each stabilizer of a isomorphic to $\text{Gal}(\mathbb{F}_{q^r}/\mathbb{F}_{q^e})$.

1.7 The Exponential and Logarithm Power Series

To understand the Hasse Weil Zeta Function, we have to first do some algebra in the ring of formal power series $\mathbb{Q}[[t]]$ with coefficients in $\mathbb{Q}$.

The key point here is that in this ring $\mathbb{Q}[[t]]$, we are only doing algebra, not analysis, therefore we don't care any numerical convergence. Given two elements $f, g \in \mathbb{Q}[[t]]$. Suppose that:

$$f = \sum_{n \geq 0} a_n t^n, \quad g = \sum_{n \geq 0} b_n t^n$$

Their addition is defined by the addition of the coefficients of the corresponding degree. Their product:

$$f \cdot g = \sum_{n \geq 0} \left(\sum_{i+j=n} a_i b_j \right) t^n$$

We define the derivative operation by:

$$f' = \left(\sum_{n \geq 0} a_n t^n \right)' = \sum_{n \geq 0} (n+1) a_{n+1} t^n$$

This derivative satisfies the usual derivative rules for example it's linear and satisfies the product rule. Essentially why this is true is because we know that when you are computing the derivatives, each term is only determined by finitely many terms in f and in g , therefore we know that we can directly apply the product rule for polynomials.

Suppose that $u(t) \in t\mathbb{Q}[[t]]$ and $f(t) \in \mathbb{Q}[[t]]$, we can define a composition $f(u(t))$ which should look like:

$$f(u(t)) = \sum_{n \geq 0} a_n u(t)^n$$

To be more precise, to make this an element in $\mathbb{Q}[[t]]$, we need to specify what is the coefficient of each degree. This is exactly why we set $u(t) \in t\mathbb{Q}[[t]]$ since we know that if we want to determine the coefficient, for say t^N , we know that the coefficients in $u(t)^{N+1}$ and in all the other terms doesn't contribute to the coefficients in degree N .

After we define this composition, we find that the derivative satisfies the composition law as well:

$$[f(u(t))]' = f'(u(t)) \cdot u'(t)$$

The reason is again that each term in $[f(u(t))]'$ are determined by only finitely many terms of f, u , therefore we can use the usual rule for polynomials.

Motivated by our knowledge from analysis, we will denote the following two power series by $\exp(t)$ and $\log(1+t)$:

$$\exp(t) = \sum_{n \geq 0} \frac{t^n}{n!}, \quad \log(1+t) = \sum_{n \geq 1} \frac{(-1)^{n+1} t^n}{n}$$

Again, we emphasize that this is just a notation and it carries no convergence assumption.

Similarly, we can consider the ring $\mathbb{Q}[[s, t]]$ which is the formal power series ring with two variables. Everything is defined similar to the ring with one variable. Composition now makes sense if $u(s, t), v(s, t)$ are in $(t, s)\mathbb{Q}[[s, t]]$. We can also compute the partial derivatives which works again similar to polynomials of several variables.

Note we have a $\mathbb{Q}$ -algebra morphism from:

$$\mathbb{Q}[[s, t]] \rightarrow \mathbb{Q}[[t]]$$

which is defined by where s, t is mapped to but note that s, t have to be mapped to something in $t\mathbb{Q}[[t]]$. It's clear how this map is defined, and we only need to make sure that this is indeed a ring map. But again, we know that this is well defined since if we want to check the product behaves the way we expect, each term is only determined by finitely many terms, thus we are back to the usual polynomial ring case. Then one can check that if $s \mapsto u(t)$ and $t \mapsto v(t)$, then $f(s, t) \mapsto f(u(t), v(t))$.

What is different to the ordinary polynomial ring is that now we **don't have** the evaluation map:

$$\mathbb{Q}[[t]] \rightarrow \mathbb{Q}$$

Remark 1.30. One really useful property in the ring $\mathbb{Q}[[t]]$ is that if we want to check that some power series is equal to another, we only need to show that the difference has 0 derivative and they share the same constant term. You can check it very easily and you might think this is not so useful, but you will see how it's used later.

Proposition 1.31. The units in $\mathbb{Q}[[t]]$ are precisely those elements in $\mathbb{Q}^* + t\mathbb{Q}[[t]]$.

Proof. Suppose that $f(t)$ has an inverse $g(t)$. Clearly the constant term of f can not be 0 since otherwise, the lowest degree term in the product $f(t)g(t)$ is degree 1. This is a contradiction to $f(t) \cdot g(t) = 1$.

Conversely, let $f(t) = \sum_{n \geq 0} a_n t^n$ be a formal power series such that it has nonzero constant term we can literally compute its inverse by setting:

$$\sum_{n \geq 0} a_n t^n \cdot \sum_{n \geq 0} b_n t^n = 1$$

Clearly we know that $b_0 = \frac{1}{a_0}$. Then since the product is 1, we know that the degree 1 term should all vanish, which implies:

$$a_0 b_1 + b_0 a_1 = 0$$

Note that b_0, a_1, a_0 are already given, then b_1 is determined. Then we also need all degree 2 terms to vanish but we know that now a_0, a_1, a_2, b_0, b_1 are given, then b_2 are computed. Then we keep this process to find the inverse. $\square$

Remark 1.32. If $g(t)$ is the inverse of $f(t)$, then we again have the following derivative rule for $g(t)$ where we get:

$$g(t)' f(t) + g(t) f'(t) = 0 \Rightarrow g'(t) = -g(t)^2 f'(t) = \frac{-f'(t)}{f(t)^2}$$

Some of the well known properties of these two power series are in the below proposition:

Proposition 1.33. 1. $\exp(x)' = \exp(x)$ and $\log(1+t)' = \frac{1}{1+t}$.

2. $\exp(s+t) = \exp(s) \cdot \exp(t) \in \mathbb{Q}[[s, t]]$. In particular, if $u(t), v(t) \in t\mathbb{Q}[[t]]$, then we have:

$$\exp(u(t) + v(t)) = \exp(u(t)) \cdot \exp(v(t))$$

3. $\exp(mt) = \exp(t)^m$ for all $m \in \mathbb{Z}$. In particular we have:

$$\exp(mu(t)) = \exp(u(t))^m, \quad u(t) \in t\mathbb{Q}[[t]], \quad m \in \mathbb{Z}$$

4. $\log(\exp(u(t))) = u(t)$ and $\exp(\log(1+u)) = 1+u$ for all $u(t) \in t\mathbb{Q}[[t]]$.

5. $\log((1+u(t))(1+v(t))) = \log(1+u) + \log(1+v)$ for all $u, v \in t\mathbb{Q}[[t]]$.

6. $\log((1+u)^m) = m \log(1+u)$ for all $m \in \mathbb{Z}$ and $u \in t\mathbb{Q}[[t]]$.

Proof. Let's only prove some of them as all the proof are all similar.

Suppose that we want to show:

$$\log(1+t)' = \frac{1}{1+t}$$

We know that by definition:

$$\log(1+t)' = 1 - t + t^2 - t^3 + \dots$$

Then we know that if we compute:

$$(1+t)(1-t+t^2-t^3+\dots)$$

Clearly by the algebraic rule, it's equal to 1.

For example, we want to show that:

$$\exp(s+t) = \exp(s) \cdot \exp(t)$$

We know that:

$$\exp(s+t) = \sum_{n \geq 0} \frac{(s+t)^n}{n!}$$

We know that if we consider what are those terms with $s^i t^j$, we know that they can only come from:

$$\frac{(s+t)^{i+j}}{(i+j)!}$$

We know that the coefficient is:

$$\frac{1}{(i+j)!} \cdot \binom{i+j}{i} = \frac{1}{(i+j)!} \cdot \frac{(i+j)!}{i! \cdot j!} = \frac{1}{i! \cdot j!}$$

This is exactly what the coefficient you get from $\exp(s) \cdot \exp(t)$.

Then recall the ring map:

$$\mathbb{Q}[[s, t]] \rightarrow \mathbb{Q}[[t]]$$

We know that if we consider the ring map which maps s to $u(t)$ and maps t to $v(t)$, then:

$$\exp(s+t) = \exp(s) \cdot \exp(t) \mapsto \exp(u(t)) \cdot \exp(v(t))$$

This proves 1, 2 and clearly 3 follows from 2.

To prove 4, consider the derivative:

$$\log(\exp(u(t)))'$$

Using the derivative property, we know that it's just:

$$\frac{1}{\exp(u)} \exp(u(t)) u'(t) = u'(t)$$

Then notice that the constant term of $\log(\exp(u(t)))$ is just the constant of $u(t)$, then we know that we are done by the above remark.

To prove:

$$\exp(\log(1+u)) = 1+u$$

Notice that $1+u(t) \in \mathbb{Q}^* + t\mathbb{Q}[[t]]$, therefore it has an inverse $h(t)$. Then we only need to prove that:

$$\exp(\log(1+u))h(t) = 1$$

Now compute it's derivative, which gives:

$$\exp(\log(1+u)) \frac{1}{1+u(t)} u'(t) h(t) + h'(t) \exp(\log(1+u))$$

I claim that:

$$h'(t) = \frac{-u'(t)}{(1+u(t))^2}$$

Then we can rewrite the derivative as:

$$\exp(\log(1+u)) \left(\frac{u'(t)}{(1+u(t))^2} - \frac{u'(t)}{(1+u(t))^2} \right) = 0$$

Then we check that the constant term of $\exp(\log(1+u))h(t) = 1$, then we are done.

Knowing 4, 5 follows easily since if $\exp(u(t)) = \exp(v(t))$ Then we can take log to conclude that $u = v$, then we can apply exp to $\log((1+u)(1+v))$ and $\log(1+u) + \log(1+v)$ which gives us:

$$(1+u)(1+v) = \exp(\log(1+u)) \cdot \exp(\log(1+v)) = (1+u)(1+v)$$

Therefore the formal power series before we apply exp must also be equal.

6 is just a special case of 5. □

1.8 The Hasse Weil Zeta Function

Now let's give the definition of the Hasse Weil Zeta Functions:

Suppose that X is a variety over a finite field $k = \mathbb{F}_q$, for every $m \geq 1$, let $N_m = |X(\mathbb{F}_{q^m})|$. To avoid confusion let's assume that $\mathbb{F}_{q^m}$ is in a given algebraic closure of $\mathbb{F}_q$ and the extension is just the canonical extension from $\mathbb{F}_q$ to $\mathbb{F}_{q^m}$. Then the Hasse-Weil zeta function of X is defined by:

$$Z(X, t) = \exp\left(\sum_{m \geq 1} \frac{N_m}{m} t^m\right) \in \mathbb{Q}[[t]]$$

Note that $\sum_{m \geq 1} \frac{N_m}{m} t^m \in t\mathbb{Q}[[t]]$, therefore this is well defined. The following gives a product formula for $Z(X, t)$ that is very useful in practice.

Proposition 1.34. For every variety X over k , we have:

$$Z(X, t) = \prod_{x \in X_{cl}} (1 - t^{\deg(x)})^{-1}$$

In particular, $Z(X, t) \in \mathbb{Z}[[t]]$.

Before we prove this proposition, let's try to make sense of what is this infinite product.

Let's first try to explain why this product is almost always infinite. Recall that if X is a variety over k , then it has only finitely many closed point iff it's 0 dimensional. Indeed, we know that closed points is dense in any locally finite type scheme over a field k , which is basically the result of Hilbert's Nullstellensatz which says that a point is closed iff it's residue class field is finite dim over k . In particular, we know that for a scheme X locally finite type over k , X has at least 1 closed point, which further implies every closed subset of X at least contains a closed point. Indeed, if we give this closed subset the canonical reduced subscheme structure, it's residue class field is not affected, and we know it still has at least a closed point, which is also closed in X .

Then we know that if X is dimension 0 then each of its points should be closed. Conversely, if X is a finite collection of points X is dim 0. This also proves ff X is not dimension 0, it has infinitely many closed points. You can also prove it by showing that if X is dim $d > 0$ and irreducible and affine, we know that using Noether's normalization, there is a finite surjective morphism:

$$\text{Spec } A \rightarrow \mathbb{A}_k^n$$

Recall that we have a closed embedding of $\mathbb{A}_k^1 \hookrightarrow \mathbb{A}_k^n$ and we know that $\mathbb{A}_k^1$ has infinitely many closed points since it has infinitely many irreducible polynomials. Therefore $\mathbb{A}_k^n$ has infinitely many points. Notice that if $x \in \mathbb{A}_k^n$ is closed, then it's preimage is closed, which contains a closed point, therefore we are done.

But either way, if X is not just a finite collection of discrete points, we know that X has infinitely many closed points, therefore we have to make sense of this product.

Note that even though X has infinitely many closed point, if we consider the degree of x , we know that there are only finitely many closed points whose degree is bounded, say at most N . Recall that if $\deg(x) = r$, then we can have a $\mathbb{F}_{q^r}$ -valued point whose image is at x . But there are only finitely many $\mathbb{F}_{q^r}$ -points, therefore there are only finitely many closed points with degree r . Then we will define this product by:

$$\prod_{d \geq 1} \prod_{x \in X_{cl}, \deg(x)=d} (1 - t^{\deg(x)})^{-1} = \prod_{d \geq 1} (1 + t^{\deg(x)} + t^{2\deg(x)} + \dots)_{|x: \deg(x)=d|}$$

Notice that the inner product is well defined since we know that there are only finitely many x with $\deg x = r$. To define an element in $\mathbb{Q}[[t]]$, we need to define the coefficient of each degree. Notice that if we want to determine the coefficient of degree N , if $\deg(x) > N$, then we know that those terms with $t^{\deg(x)} > N$ doesn't affect the coefficient of degree N . Therefore we know that to determine the coefficient, we only need to consider finitely many terms in the product. This is how we understand this product. Then notice that each $(1 - t^{\deg(x)})^{-1}$ is in $\mathbb{Z}[[t]]$, then infinite product is also in $\mathbb{Z}[[t]]$.

Then let's prove this proposition:

Proof. Let:

$$a_r = |\{x \in X_{\text{cl}} : \deg(x) = r\}|$$

Then we know that the infinite product is:

$$\prod_{r \geq 1} (1 - t^r)^{-a_r}$$

Recall that:

$$N_m = |X(\mathbb{F}_{q^m})| = \prod_{r|m} r \cdot a_r$$

Now apply log to $Z(X, t)$ which gives us:

$$\log Z(X, t) = \sum_{m \geq 1} \frac{N_m}{m} t^m = \sum_{m \geq 1} \left(\sum_{r|m} \frac{r \cdot a_r}{m} t^m \right)$$

Notice that the element in $\mathbb{Q}[[t]]$ defined by this summation can also be thought as the following summation:

$$\sum_{m \geq 1} \sum_{r|m} \frac{a_r}{\frac{m}{r}} t^{\frac{m}{r} \cdot r}$$

Therefore we know that if we want to determine the coefficient of degree m , we first find out all the divisor r of m , then consider what is a_r , then divide a_r by s where $s = m/r$. Then consider the following element:

$$\sum_{r \geq 1} a_r \sum_{\ell \geq 1} \frac{t^{\ell \cdot r}}{\ell}$$

I will let you find that this summation defines the same element as above.

Therefore we know that:

$$\log Z(X, t) = \sum_{r \geq 1} a_r \sum_{\ell \geq 1} \frac{t^{\ell \cdot r}}{\ell} = \sum_{r \geq 1} -a_r \log(1 - t^r)$$

Again, I will let you check this infinite summation of power series is well defined. Moreover, I claim we have:

$$\exp\left(\sum_{r \geq 1} -a_r \log(1 - t^r)\right) = \prod_{r \geq 1} \exp(-a_r \log(1 - t^r)) = \prod_{r \geq 1} (1 - t^r)^{-a_r}$$

Indeed, even if we apply exp to an infinite sum, we know that to determine each coefficient, since only finitely many terms are relevant, we can think of this as a finite sum.

This shows:

$$Z(X, t) = \prod_{r \geq 1} (1 - t^r)^{-a_r}$$

as desired. □

Remark 1.35. This notion of zeta function is very much dependent on which base field we are working with. Suppose that $q = (q')^m$, then we know that:

$$Z(X/\mathbb{F}_q, t) = \prod_{x \in X_{\text{cl}}} (1 - t^{\deg(x)})^{-1}$$

If now we think of X as over $\mathbb{F}_{q'}$, clearly this doesn't change where x is a closed or not. However, we know that the degree of x over $\mathbb{F}_{q'}$ is m times the degree of x over $\mathbb{F}_q$, therefore:

$$Z(X/\mathbb{F}_{q'}, t) = \prod_{x \in X_{\text{cl}}} (1 - t^{\deg(x)m})^{-1} = Z(X/\mathbb{F}_q, t^m)$$

Remark 1.36. Recall that for a variety X over k , the group of 0 cycles is the free abelian group generated by all its closed points. We say that a 0-cycle is effective if all the coefficients are nonnegative. Given a 0-cycle:

$$\alpha = \sum_{i=1}^r m_i x_i$$

We define its degree to be $\deg(\alpha) = \sum_i m_i \deg(x_i)$. Recall that we can write:

$$Z(X, t) = \prod_{x \in X_{\text{cl}}} (1 + t^{\deg(x)} + t^{2\deg(x)} + \dots)$$

We can also consider the following summation:

$$\sum_{\alpha} t^{\deg(\alpha)}$$

where α ranges over all effective 0-cycles of X .

I claim that these two define the same element. Indeed, if we want to find the coefficient of degree N . In the first term, we will consider all x such that $\deg(x) \leq N$, say $x_1, \dots, x_n$, then we know that the coefficient is determined by the number of elements in $\mathbb{Z}_{\geq 0}^n$ such that:

$$\sum_i m_i \deg(x_i) = N$$

But notice that this number is equal to the number of α whose degree is N .

Let's see some examples and properties of Hasse-Weil Zeta functions.

Example 1.37. Let $k = \mathbb{F}_q$ and let $X = \mathbb{A}_k^n$. Recall that for the affine space, we know:

$$|X(\mathbb{F}_{q^m})| = |\mathbb{F}_{q^m}|^n = q^{mn}$$

Therefore by definition:

$$Z(X, \mathbb{F}_q) = \exp\left(\sum_{m \geq 1} \frac{q^{mn}}{m} t^m\right) = \exp(-\log(1 - q^n t)) = \frac{1}{1 - q^n t}$$

More generally, recall that for two k -varieties X, Y , we always have:

$$X \times_k Y(\mathbb{F}_{q^m}) = X(\mathbb{F}_{q^m}) \times Y(\mathbb{F}_{q^m})$$

This implies that:

$$Z(X \times_k Y, t) = \exp\left(\sum_{m \geq 1} \frac{N_{m,X} N_{m,Y}}{m} t^m\right) = \exp\left(\sum_{m \geq 1} \frac{N_{m,X}}{m} t^m + \sum_{m \geq 1} \frac{N_{m,Y}}{m} t^m\right)$$

Then if $X = \mathbb{A}_k^n$, we know that the above formula can be written as:

$$\exp\left(\sum_{m \geq 1} \frac{q^{mn} N_{m,Y}}{m} t^m\right) = \exp\left(\sum_{m \geq 1} \frac{N_{m,Y}}{m} (q^n t)^m\right) = Z(Y, q^n t)$$

Notice that when $n = 0$, we know that:

$$Z(\mathbb{A}_k^0, t) = \exp\left(\sum_{m \geq 1} t^m\right) = \frac{1}{1 - t}$$

Then using the above result, since $\mathbb{A}_k^n = \mathbb{A}_k^0 \times_k \mathbb{A}_k^n$, we get:

$$Z(\mathbb{A}_k^n, t) = Z(\mathbb{A}_k^0, q^n t) = \frac{1}{1 - q^n t}$$

Proposition 1.38. Let X be a variety over $\mathbb{F}_q$ and let $Y \hookrightarrow X$ be a closed subvariety of X , then:

$$Z(X, t) = Z(Y, t) \cdot Z(U, t)$$

where $U = X \setminus Y$.

Proof. The key is to notice that if $x \in X(\mathbb{F}_{q^m})$ is a $\mathbb{F}_{q^m}$ point, then consider its image x , it's either in Y or in U . Then suppose that it's in Y , we have the following unique factorization:

$$\begin{array}{ccc} & \text{Spec} \mathbb{F}_{q^m} & \\ \swarrow & & \searrow \\ Y & \xrightarrow{\quad} & X \end{array}$$

Indeed, the residue class field of Y at x is the same as the residue class field of X at x , therefore we find the existence of such a factorization easily.

This factorization is unique because any such factorization gives:

$$\begin{array}{ccc} & \mathbb{F}_{q^m} & \\ \nearrow & & \nwarrow \\ \mathcal{O}_{Y,x} & \xleftarrow{\quad} & \mathcal{O}_{X,x} \end{array}$$

Since $\mathcal{O}_{Y,x}$ is a quotient of $\mathcal{O}_{X,x}$, the map from $\mathcal{O}_{Y,x}$ to $\mathbb{F}_{q^m}$ is forced. This implies that we can identify $Y(\mathbb{F}_{q^m})$ as a subset of $X(\mathbb{F}_{q^m})$.

Then we know that if the image of x is in U , then this is automatically a point in $U(\mathbb{F}_{q^m})$. The factorization is also unique. This implies that we can identify $U(\mathbb{F}_{q^m})$ as a subset of $X(\mathbb{F}_{q^m})$. Therefore we have:

$$X(\mathbb{F}_{q^m}) = Y(\mathbb{F}_{q^m}) \sqcup U(\mathbb{F}_{q^m})$$

Then we know that:

$$\begin{aligned} \exp\left(\sum_{m \geq 1} (|X(\mathbb{F}_{q^m})|/m)t^m\right) &= \exp\left(\sum_{m \geq 1} \frac{|Y(\mathbb{F}_{q^m})|}{m}t^m + \sum_{m \geq 1} \frac{|U(\mathbb{F}_{q^m})|}{m}t^m\right) \\ &= \exp\left(\frac{|Y(\mathbb{F}_{q^m})|}{m}t^m\right) \cdot \exp\left(\frac{|U(\mathbb{F}_{q^m})|}{m}t^m\right) = Z(Y, t) \cdot Z(U, t) \end{aligned}$$

□

Corollary 1.39. The zeta function of the projective space is given by:

$$Z(\mathbb{P}_{\mathbb{F}_q}^n, t) = \frac{1}{(1-t)(1-qt) \cdots (1-q^n t)}$$

Proof. Recall that we have a closed embedding:

$$\mathbb{P}_k^{n-1} \hookrightarrow \mathbb{P}_k^n$$

defined by the map of graded rings:

$$k[x_0, \dots, x_n] \rightarrow k[x_0, \dots, x_{n-1}], \quad x_i \mapsto x_i, \quad x_n \mapsto 0$$

Recall that $\mathbb{A}^n \simeq D_+(x_n) \subset \mathbb{P}_k^n$ has empty preimage and the image of this map is just $V(x_0)$. Therefore we know that by the previous proposition:

$$Z(\mathbb{P}_{\mathbb{F}_q}^n, t) = Z(\mathbb{P}_{\mathbb{F}_q}^{n-1}, t) \cdot \mathbb{A}_k^n = Z(\mathbb{P}_{\mathbb{F}_q}^{n-1}, t) \cdot \frac{1}{1-q^n t}$$

Then we can use induction to get the desired result.

□

Proposition 1.40. Let X be a variety over $k = \mathbb{F}_q$ and let k'/k be a field extension of degree r . If $X' = X \times_k k'$, then:

$$Z(X', t^r) = \prod_{i=1}^r Z(X, \xi_i t)$$

where ξ_i is a primitive r -th root of unity.

Proof. Let $k' = \mathbb{F}_{q^r}$. Then we will let:

$$N'_m = |X'(\mathbb{F}_{q^{rm}})|, \quad N_m = |X(\mathbb{F}_{q^m})|$$

Recall that:

$$X'(\mathbb{F}_{q^{rm}}) \simeq X(\mathbb{F}_{q^{rm}})$$

Then we know that:

$$N'_m = N_{mr}$$

Then by definition it's enough to show that:

$$\sum_{m \geq 1} \frac{N_{mr}}{m} t^{mr} = \sum_{i=1}^r \sum_{\ell \geq 1} \frac{N_\ell}{\ell} \xi^{i \cdot \ell} t^\ell = \sum_{\ell \geq 1} \frac{N_\ell}{\ell} \sum_{i=1}^r \xi^{i \ell} t^\ell$$

Then we can apply exp to both sides and get the desired equality.

Note that if $r \nmid \ell$, then ξ^ℓ is still a r -th root of unity, though may not be primitive is not equal to 1. Indeed, if $\xi^\ell = 1$, then we know that r has to divide ℓ . Then in this case I will let you show that:

$$\sum_{i=1}^r \xi^{i \cdot \ell} = 0$$

If $r \mid \ell$, then we know that in this case:

$$\sum_{i=1}^r \xi^{i \cdot r} = r$$

Therefore we know that:

$$\sum_{\ell \geq 1} \frac{N_\ell}{\ell} \sum_{i=1}^r \xi^{i \ell} t^\ell = \sum_{m \geq 1} \frac{N_{mr}}{mr} r t^{mr} = \sum_{m \geq 1} \frac{N_{mr}}{m} t^{mr}$$

□

1.9 Statement of Weil Conjectures

Suppose that X is a smooth geometrically connected and projective variety of dimension n defined over a finite field $k = \mathbb{F}_q$. We will in this section use $Z(t)$ to denote $Z(X, t)$.

Conjecture 1.41. (Rationality) $Z(t)$ is a function in $\mathbb{Q}(t)$.

Conjecture 1.42. (Functional equation) Let $E = (\Delta^2)$ be the self-intersection of the diagonal $X \times_k X$, then:

$$Z\left(\frac{1}{q^n t}\right) = \pm q^{\frac{nE}{2}} t^E Z(t)$$

Conjecture 1.43. (Analogue of Riemann Hypothesis) One can write:

$$Z(t) = \frac{P_1(t) \cdot P_3(t) \cdots P_{2n-1}(t)}{P_0(t) \cdot P_2(t) \cdots P_{2n}(t)}$$

where recall that n is the dimension of X .

Moreover we have $P_0(t) = 1 - t$ and $P_{2n}(t) = 1 - q^n t$ and for $1 \leq 2n - 1$ we have $P_i(t) \in \mathbb{Z}[t]$ and we have a decomposition:

$$P_i(t) = \prod_j (1 - \alpha_{ij} t)$$

where α_{ij} are algebraic integers with $|\alpha_{ij}| = q^{i/2}$.

Remark 1.44. Suppose that there are two sets of polynomials in $\mathbb{Z}[t]$:

$$P_0(t), \dots, P_{2n}(t), \quad Q_0(t), \dots, Q_{2n}(t)$$

Such that:

$$Z(t) = \frac{P_1(t) \cdot P_3(t) \cdots P_{2n-1}(t)}{P_0(t) \cdot P_2(t) \cdots P_{2n}(t)}, \quad Z(t) = \frac{Q_1(t) \cdot Q_3(t) \cdots Q_{2n-1}(t)}{Q_0(t) \cdot Q_2(t) \cdots Q_{2n}(t)}$$

such that we have decompositions:

$$P_i(t) = \prod_j (1 - \alpha_{ij}t), \quad Q_i = \prod_j (1 - \beta_{ij}t)$$

with $|\alpha_{ij}| = |\beta_{ij}| = q^{i/2}$.

Then I claim that for all i , $P_i(t) = Q_i(t)$. Indeed, we know that if β is a root of $P_i(t)$ with multiplicity m , then $|\beta| = q^{-i/2}$. Notice that if $i \neq j$, then we know that any zeros of $P_j(t)$ is of absolute value $q^{-j/2} \neq q^{-i/2}$, therefore we know that if i is even, $Z(t)$ as a function in $\mathbb{Q}(t)$ has a pole at β of order m and this doesn't get canceled by the product on the numerator since the roots of the factors in the numerator also has different absolute value. Therefore we know that $P_i(t)$'s determine the zeros and poles of $Z(t)$ without cancellation.

Similarly, we can run the same argument for Q_i and this gives another set of zeros and pole of $Z(t)$. But we know that they should be the same the set since they are zeros and poles of the same function. Then you can prove $P_i(t) = Q_i(t)$ since they have the same zeros and they are both of the form $\prod_j (1 - \alpha_{ij}t)$.

Finally, the last part of Weil's Conjecture is:

Conjecture 1.45. (Betti-Numbers) If we assume the above conjecture on the decomposition of $Z(t)$, then we define the i -th Betti number of X to be:

$$b_i(X) = \deg(P_i(t))$$

In the case the followings hold:

1. $E = \sum_{i=0}^{2n} (-1)^i b_i(X)$.
2. Suppose that R is a finitely generated $\mathbb{Z}$ -algebra of the field $\mathbb{C}$ of complex numbers together with $\tilde{X}$ a smooth projective scheme over $\text{Spec} R$ with $P \in \text{Spec} R$ such that $R/P = \mathbb{F}_q$ and $\tilde{X} \times_R \mathbb{F}_q = X$, then:

$$b_i(X) = \dim_{\mathbb{Q}} H^i((\tilde{X} \times_R \mathbb{C})^{\text{an}}, \mathbb{Q})$$

Remark 1.46. In fact, we can see later that we are able to formulate the Betti-Numbers Conjecture without assuming Conjecture on the Riemann Hypothesis. And in the next chapter we will give the proof of the above 4 conjectures in the case of curves.

To prove the more general conjecture, we will need to use ℓ -adic cohomology which is far beyond the scope of this notes. But if we only want to prove the rationality conjecture, a proof using analysis on p -adic numbers is available, and we will give it in the end after we are finished with the conjectures of curves.

1.10 Some Remark on the Conjectures

2 Weil Conjectures of Curves

2.1 Excursion on Irreducibility

Before we proceed to prove the Weil Conjectures for curves. Recall that if X is a k -scheme then clearly we know that if $X_K = X \times_k K$ is irreducible for all extensions K/k it implies that X_K is irreducible for those algebraically closed field extension K/k . This further implies that $X_{\bar{k}}$ is irreducible. This will imply X_{k^s} where k^s is the separable closure of k is irreducible. The last one is not that trivial but clearly we can assume that X is affine and say is $\text{Spec} A$, then we only need to see that:

$$A \otimes_k \bar{k}^s \hookrightarrow A \otimes_k \bar{k}$$

is a subalgebra. Notice that if we let $B = A \otimes_k \bar{k}^s$ and $C = A \otimes_k \bar{k}$, then we know that tensor any B -module with C is the same as tensor over k with $\bar{k}$, therefore this is a faithfully flat map of rings, which means that the map induced on spectra is surjective, then we can use the image of an irreducible subset is irreducible to conclude that $X \otimes_k \bar{k}$ is irreducible.

Our goal in this short excursion is to prove that if we know that $X \otimes_k k^s$ is irreducible, then $X \otimes_k K$ is irreducible for any field extension K . To prove this, we will need a new notion of morphisms.

Definition 2.1. Suppose that $\pi : X \rightarrow Y$ is a morphism of schemes, then it's called radical or **universally injective** if it's injective under any base change of Y .

For example, you can try to show that all monomorphisms are universally injective by first show that monomorphisms are injective and being monomorphisms are preserved by all base change.

Now suppose we want to determine whether a given morphism is universally injective or not. We know that a map of sets is injective iff every fiber contains at most 1 element. Recall that we have the following Cartesian diagram about fibers:

$$\begin{array}{ccccc}
 & & \tau^{-1}(z) & \longrightarrow & \mathrm{Spec} k(z) \\
 & \swarrow & \downarrow & \swarrow & \downarrow \\
 \pi^{-1}(y) & \longrightarrow & \mathrm{Spec} k(y) & & \\
 \downarrow & & \downarrow & & \downarrow \\
 X & \xrightarrow{\pi} & Y & \xrightarrow{f} & Z \\
 & \nwarrow g & \uparrow \tau & \nwarrow & \\
 & & W & \xrightarrow{\tau} & Z
 \end{array}$$

Then we know that the top square is again Cartesian, therefore we know that the fiber of the base change is the base change of the fiber. You will very soon see that to make sure $W \rightarrow Z$ is injective, it's very much just a question of field theory. Moreover, you will also see why universally injective is the algebro-geometric generalization of **purely-inseparable extensions**.

Proposition 2.2. Suppose that E/F is a field extension with morphism of schemes $\phi : \mathrm{Spec} E \rightarrow \mathrm{Spec} F$. Then if E contains a **transcendental** element over F , then ϕ is not universally injective.

Proof. Consider the following diagram:

$$\begin{array}{ccccc}
 \mathrm{Spec} E \otimes_F F(s) & \longrightarrow & \mathrm{Spec} F(t) \otimes_F F(s) & \longrightarrow & \mathrm{Spec} F(s) \\
 \downarrow & & \downarrow & & \downarrow \\
 \mathrm{Spec} E & \longrightarrow & \mathrm{Spec} F(t) & \longrightarrow & \mathrm{Spec} F
 \end{array}$$

Notice that here t, s are independent variables over F , and the map $F(t) \rightarrow E$ is defined by mapping t to x . Then recall that:

$$F(t) \otimes_F F(s)$$

is actually a localization of $F[t, s]$ at the product of all polynomial $f(t)g(s)$.

Note that $E \otimes_F F(s)$ is in general not a field since we know that surjectivity is preserved by base change, thus $\mathrm{Spec} E \otimes_F F(s) \rightarrow \mathrm{Spec} F(t) \otimes_F F(s)$ is surjective and we know that there are at least two points in $\mathrm{Spec} F(t) \otimes_F F(s)$. For example (0) and the one corresponding to $(s + t)$. However, $\mathrm{Spec} F(s)$ is just a single point, therefore we know that at least 2 points in $\mathrm{Spec} E \otimes_F F(s)$ is mapped to the same point in $\mathrm{Spec} F(s)$, not injective. $\square$

Proposition 2.3. Using the notation from the above proposition, if E contains an algebraic element x that is **separable** over F , then ϕ is not universally injective.

Proof. Let m_x be the minimal polynomial of x over F , then consider the following diagram:

$$\begin{array}{ccccc} \text{Spec?} & \longrightarrow & \text{Spec}F[t]/(m_x(t)) \otimes_F \overline{F} & \longrightarrow & \text{Spec}\overline{F} \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec}E & \longrightarrow & \text{Spec}F[t]/(m_x(t)) & \longrightarrow & \text{Spec}F \end{array}$$

I am using a question mark ? on the top left corner because we don't have to know what is it. Clearly, let's assume that the degree of m_x is at least 2 otherwise this element x is just in F itself. But if this is true, then we know that since m_x splits into distinct linear factors in $\overline{F}$, then using the Chinese Remainder Theorem, we know that $\text{Spec}F[t]/(m_x(t)) \otimes_F \overline{F}$ is at least 2 points, therefore again $\text{Spec?} \rightarrow \text{Spec}F[t]/(m_x(t)) \otimes_F \overline{F}$ is surjective, which means that $\text{Spec?} \rightarrow \text{Spec}\overline{F}$ is not injective. $\square$

Finally:

Proposition 2.4. Suppose that E/F contains no element of the above form, which means that E/F is algebraic and is **purely inseparable**. Then if E'/F is any other field extension, $\text{Spec}E \otimes_F E'$ contains precisely one element.

Proof. Suppose that E/F is purely inseparable, which means that for every $x \in E$, there is a N such that $x^{p^N} \in F$. Then consider the ring map:

$$E' \rightarrow E' \otimes_F E$$

First, we notice that we can assume that E' is always algebraically closed since we will have:

$$\begin{array}{ccccc} \text{Spec}\overline{E'} \otimes_F E & \longrightarrow & \text{Spec}E' \otimes_F E & \longrightarrow & \text{Spec}E \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec}\overline{E'} & \longrightarrow & \text{Spec}E' & \longrightarrow & \text{Spec}F \end{array}$$

Notice if we assume there are nontrivial purely inseparable field extensions, then the characteristic of these fields has to be p .

Then I will let you check that for any element y in $E' \otimes_F E$, there is a N such that $y^{p^N} \in E'$. Then let $\mathfrak{N}$ be the nilradical of $E' \otimes_F E$, I claim that the quotient:

$$(E' \otimes_F E)/\mathfrak{N}$$

is a field. Indeed, we still have a map:

$$E' \rightarrow E' \otimes_F E \rightarrow (E' \otimes_F E)/\mathfrak{N}$$

Moreover, for every nonzero element y in this quotient, there is still some N such that $y^{p^N} \in E'$, which means that there is an element $x \in E'$ such that $xy^{p^N} = 1 \in E$ which means that it's 1 in $(E' \otimes_F E)/\mathfrak{N}$. Since E' is algebraically closed, we know that there is x' such that $x = x'^{p^N}$, therefore we have:

$$(x'y)^{p^N} = 1 \Rightarrow x'y$$

is just a root of unity, therefore y is invertible, therefore $(E' \otimes_F E)/\mathfrak{N}$ is a field. Or you can use reducedness more explicitly by noticing:

$$(x'y - 1)^{p^N} = 0$$

Since $(E' \otimes_F E)/\mathfrak{N}$ is reduced, this implies $x'y = 1$, we are done. $\square$

Proposition 2.5. Suppose that $\pi : X \rightarrow Y$ is a morphism of schemes, the following are equivalent:

1. The morphism π is universally injective.
2. The morphism π is injective and for each $p \in X$, the field extension $\kappa(p)/\kappa(\pi(p))$ is purely inseparable

Proof. Let's first prove $2 \Rightarrow 1$: Recall that above Cartesian cube we drew, and the top square is:

$$\begin{array}{ccc} \tau^{-1}(z) & \longrightarrow & \text{Spec}\kappa(z) \\ \downarrow & & \downarrow \\ \pi^{-1}(y) & \longrightarrow & \text{Spec}\kappa(y) \end{array}$$

Notice that we know that if $\pi^{-1}y$ is empty, then so is $\tau^{-1}z$, therefore let's assume that $\pi^{-1}y$ contains just 1 element. It's affine and thus we can consider the following diagram:

$$\begin{array}{ccccc} ? & \longrightarrow & \tau^{-1}(z) & \longrightarrow & \text{Spec}\kappa(z) \\ \downarrow & & \downarrow & & \downarrow \\ \pi^{-1}(y)_{\text{red}} & \hookrightarrow & \pi^{-1}(y) & \longrightarrow & \text{Spec}\kappa(y) \end{array}$$

Notice that $\pi^{-1}(y)_{\text{red}}$ is just $\kappa(x)$, which is purely inseparable over $\kappa(y)$, therefore we have $?$ is only a single element. Since $? \rightarrow \tau^{-1}(z)$ is surjective, then $\tau^{-1}(z)$ also contains a single element.

To show $1 \Rightarrow 2$: Notice that if $\kappa(x)/\kappa(y)$ is not purely inseparable, either it's not algebraic or contains separable elements, we consider:

$$\begin{array}{ccc} ? & \longrightarrow & \text{Spec}\kappa(y)(s) \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array} \quad \begin{array}{ccc} ? & \longrightarrow & \text{Spec}\overline{\kappa(y)} \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

Then we know that in either case, the top square in the cube looks like:

$$\begin{array}{ccc} ? & \longrightarrow & \text{Spec}\kappa(y)(s) \\ \downarrow & & \downarrow \\ \pi^{-1}(y) & \longrightarrow & \text{Spec}\kappa(y) \end{array} \quad \begin{array}{ccc} ? & \longrightarrow & \text{Spec}\overline{\kappa(y)} \\ \downarrow & & \downarrow \\ \pi^{-1}(y) & \longrightarrow & \text{Spec}\kappa(y) \end{array}$$

But we know from universally injective that $\pi^{-1}y$ is a singleton, which means that we can consider the reduced subscheme structure which gives:

$$\begin{array}{ccccc} ? & \longrightarrow & ? & \longrightarrow & \text{Spec}\kappa(y)(s) \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec}\kappa(x) & \longrightarrow & \pi^{-1}(y) & \longrightarrow & \text{Spec}\kappa(y) \end{array} \quad \begin{array}{ccccc} ? & \longrightarrow & ? & \longrightarrow & \text{Spec}\overline{\kappa(y)} \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec}\kappa(x) & \longrightarrow & \pi^{-1}(y) & \longrightarrow & \text{Spec}\kappa(y) \end{array}$$

Notice that since $\text{Spec}\kappa(x) \rightarrow \pi^{-1}(y)$ is a closed embedding, this implies that $? \rightarrow ?$ is also a closed embedding, in particular injective. Recall that we have proved that $?$ on the top left corner has at least 2 points, which means that the $?$ in the middle has at least 2 points. This is a contradiction to π being universally injective which means any base change is injective and $? \rightarrow \text{Spec}\kappa(y)(s)$ or $? \rightarrow \text{Spec}\kappa(y)$ are base changes of π . $\square$

We will also need the following result which I will refer you to go to FOAG series for a proof:

Theorem 2.6.

Proposition 2.7. Suppose that k is separably closed and A is an k -algebra with $\text{Spec}A$ irreducible, then for any field extension K/k , $\text{Spec}A \otimes_k K$ is irreducible.

Proof.

$\square$

2.2 Rationality of Zeta Functions

Now we begin to prove Weil's Conjecture for curves. In this section, we will work with smooth, projective curves defined over $k = \mathbb{F}_q$. Let $\bar{k}$ be the algebraic closure of k and let $\bar{X} = X \times_k \bar{k}$ be the base change with the natural projection $\pi : \bar{X} \rightarrow X$.

Recall that since we can compute cohomology using Čech cohomology in this case, we have isomorphisms for each quasicoherent sheaf $\mathcal{F}$ on X :

$$H^i(X, \mathcal{F}) \otimes_k \bar{k} = H^i(\bar{X}, \pi^* \mathcal{F})$$

Let's also assume that X is geometrically connected which is equivalent to $\bar{X}$ is connected. Since in our setting, smoothness is equivalent to being regular and X is clearly Noetherian, connectedness is equal to irreducibility, which means that $\bar{X}$ is a smooth, irreducible projective curve over $\bar{k}$. Moreover, recall that functions on reduced proper $\bar{k}$ -schemes must be constant, therefore we have:

$$H^0(\bar{X}, \mathcal{O}_{\bar{X}}) = \bar{k} \Rightarrow H^0(X, \mathcal{O}_X) = k$$

Recall that the genus, or the arithmetic genus g of X is $h^1(X, \mathcal{O}_X) = h^1(\bar{X}, \mathcal{O}_{\bar{X}})$.

Let's also recall some basics on Weil Divisor and Line bundles in this case. Recall that since k is perfect, regular is the same as smoothness. Therefore we know that X is of course a normal scheme, then a lot of the discussions in the appendix can be utilized. Moreover, recall that X is a curve, therefore we know that normal is equivalent to being factorial, then we have that each Weil divisor D on X is locally principal, and sometimes people just call it Cartier. In this case we have an isomorphism:

$$\text{Pic}X \rightarrow \text{Cl}X$$

We will call the degree of a line bundle:

$$\deg D = \sum_i a_i [k(P_i) : k]$$

The line bundle associated to D will be denoted by $\mathcal{O}_X(D)$. Note that this degree map is clearly a group homomorphism from $\text{Cl}X$ to $\mathbb{Z}$. Indeed, if we have a principal Weil divisor which is defined by some rational function f , they it has the same number of zeros and poles counting multiplicity. This implies that its degree is 0.

This implies that we actually have a group homomorphism:

$$\deg : \text{Pic}X \rightarrow \text{Cl}X \rightarrow \mathbb{Z}$$

To compute the degree of a line bundle, we know that we only need to take a rational section of $\mathcal{L}$ and compute the degree of the divisor determined by the rational section.

Now given a line bundle $\mathcal{L}$ on X , recall that the set of effective divisors D on X whose associated line bundle is $\mathcal{L}$ is in one to one correspondence to $H^0(X, \mathcal{L}) \setminus \{0\}$ by the action of multiplication by invertible elements in $H^0(X, \mathcal{O}_X)$. Indeed this correspondence is given by fixing $\mathcal{L}$ and consider the map:

$$\text{div} : (H^0(X, \mathcal{L}) \setminus \{0\}) / H^0(X, \mathcal{O}_X)^\times \rightarrow \text{Weil}X$$

Therefore we can conclude that there is no effective divisor D whose associated line bundle is $\mathcal{L}$ iff $H^0(X, \mathcal{L}) = 0$.

We have a bijection:

$$(H^0(X, \mathcal{L}) \setminus \{0\}) / H^0(X, \mathcal{O}_X)^\times \simeq \mathbb{P}^{h^0(\mathcal{L})-1}(\mathbb{F}_q)$$

And we know that it has $\frac{q^{h^0(\mathcal{L})}-1}{q-1}$ points.

Finally, recall that we have the **Riemann-Roch theorem**:

$$\chi(X, \mathcal{O}_X(D)) = \deg(D) - g + 1$$

This is the version which involves **Serre Duality** which is unlike other versions of Riemann-Roch, this is very hard to prove.

Recall that when $\deg(\mathcal{O}_X(D)) \geq 2g - 1$, then $H^1(X, \mathcal{O}_X(D)) = 0$ which implies:

$$h^0(X, \mathcal{O}_X(D)) = \deg(D) - g + 1$$

Recall that this result comes from Serre Duality which says that:

$$H^1(X, \mathcal{O}_X(D))^\vee \simeq H^0(X, \omega_X \otimes \mathcal{O}_X(-D))$$

Recall that the degree of ω_X on X is $2g - 2$, therefore if $\deg(\mathcal{O}_X(D)) \geq 2g - 1$, then the degree of $\omega_X \otimes \mathcal{O}_X(-D)$ is negative, thus it has no global sections.

Recall that Serre duality on X says in general that if we consider the canonical bundle $\omega_X = \Omega_{X/k}$ over X , then for every line bundle $\mathcal{L}$ on X , we have:

$$h^1(X, \mathcal{L}) = h^0(X, \omega_X \otimes \mathcal{L}^\vee)$$

Finally, we will need the following result that we are not able to prove in a first course of algebraic geometry. Associated to X , there is a variety denoted by $J(X)$ which is usually called the **Jacobian variety** of X which has the following property. For each field extension k'/k , the k' -valued points of $J(X)$ are in natural bijection with the degree 0-line bundles on $X \times_k k'$, up to isomorphism.

In particular, the number h of line bundles on X of degree 0 is equal to $|J(X)(k)|$ and hence it's finite and of course positive. Then recall that we have a bijective map:

$$\otimes \mathcal{L} : \text{Pic}^0(X) \rightarrow \text{Pic}^m(X)$$

if there is at least one $\mathcal{L}$, a degree m -line bundle. Then we know that if the set degree 0 line bundles is not empty, then the number of degree m line bundles is equal to the number of degree 0 line bundles, which is h . We will later see that this is never empty, thus always there are m of them.

Now let's prove the rationality conjecture, which is the following more precise statement:

Theorem 2.8. If X is a smooth, geometrically connected projective curve of genus g over $k = \mathbb{F}_q$, then we have:

$$Z(X, t) = \frac{f(t)}{(1-t)(1-qt)}$$

where $f(t) \in \mathbb{Z}[t]$ is a polynomial of degree $\leq 2g$ with $f(0) = 1$ and $f(1) = h$ which is the number of degree 0 line bundles.

3 Appendix

3.1 Basic Intersection Theory

Throughout this section we will follow [Vak25] and probably add more details. Let X be a variety over k and for most of the application, we will let X be projective.

Recall that given some line bundles $\mathcal{L}_1, \dots, \mathcal{L}_n$ on X , we can try to define it's intersection product with another **coherent sheaf** $\mathcal{F}$ whose support is at most of dimension n , which is denoted by:

$$(\mathcal{L}_1, \dots, \mathcal{L}_n, \mathcal{F})$$

And this notation is defined by:

$$\sum_{\{i_1, \dots, i_m\} \subset \{1, \dots, n\}} (-1)^m \chi(X, \mathcal{L}_{i_1}^\vee \otimes \dots \otimes \mathcal{L}_{i_m}^\vee, \mathcal{F})$$

There are several questions here:

1. Why we are only defining it for coherent sheaves.

2. Why we require the support of $\mathcal{F}$ to have at most dimension n and proper
3. What exactly do we mean by the support.

First, we are working with varieties so coherent is equivalent to finite type quasicoherent. Recall that the support of a finitely type quasicoherent sheaf is always closed which is not true for an arbitrary $\mathcal{O}_X$ -module (if you remember the support is defined to be the locus x where $\mathcal{F}_x \neq 0$).

Then the reason we want the support to be proper of dimension at most n will be explained later which basically is some finiteness assumptions.

Finally, let's explain what we mean by the support of $\mathcal{F}$. Recall that given any closed subset of a scheme, we can put the reduced closed subscheme structure on it. Since $\text{Supp}\mathcal{F}$ is closed, we can first think of it as a reduced closed subscheme:

$$(\text{Supp}\mathcal{F})_{\text{red}} \hookrightarrow X$$

However, there is another notion of the support of a coherent sheaf. Recall that we can define the annihilator ideal sheaf:

$$\text{Ann}\mathcal{F} \hookrightarrow \mathcal{O}_X$$

This is defined by on each $U \subset X$:

$$\text{Ann}\mathcal{F}(U) = \{f \in \mathcal{O}_X(U) : f_x \mathcal{F}_x = 0, \forall x \in U\}$$

Geometrically speaking, you can think of this as some the local functions that completely kills $\mathcal{F}|_U$. One common mistake when defining this is to make the definition of $f\mathcal{F}(U) = 0$. This is not correct because remember sheaf is essentially a local data and to kill a sheaf is to kill all it's local data which means you need to kill all stalks.

Then you can check that this is indeed a presubsheaf of $\mathcal{O}_X$ and the gluing condition is also automatically inherited from that on $\mathcal{O}_X$, which makes is an ideal sheaf.

What is important is that when $\mathcal{F}$ is coherent, this $\text{Ann}\mathcal{F}$ is quasicoherent. Indeed, you can check that on $\text{Spec}A$, the sections of $\text{Ann}\mathcal{F}(\text{Spec}A)$ are defined to be the annihilator of $\mathcal{F}(\text{Spec}A)$ as an A -module. When $\mathcal{F}$ is coherent, we know that this construction is compatible with localizations, therefore makes it quasicoherent.

Then we can consider the closed subscheme defined by this quasicoherent ideal sheaf $\text{Ann}\mathcal{F}$, which is denoted by $\text{Supp}\mathcal{F} \hookrightarrow X$. Notice that the underlying set of this closed subscheme is still $\text{Supp}\mathcal{F}$ and you can check this locally.

What is interesting is that the ideal sheaf defining $\text{Ann}\mathcal{F}$ is actually a subideal sheaf of that defining $(\text{Ann}\mathcal{F})_{\text{red}}$. Indeed, we see that on each affine $\text{Spec}A$, the ideal sheaf defining $(\text{Ann}\mathcal{F})_{\text{red}}$ is just $\sqrt{\text{Ann}\mathcal{F}(U)}$. This gives us a canonical closed embedding:

$$\begin{array}{ccc} (\text{Ann}\mathcal{F})_{\text{red}} & \hookrightarrow & \text{Ann}\mathcal{F} \\ & \searrow & \swarrow \\ & X & \end{array}$$

Notice that no matter it's reduced or not, the two closed subschemes have the same homeomorphic underlying topological space.

Then by the dimension of the $\text{Supp}\mathcal{F}$ is at most n , we can say it for either the underlying topological space of $(\text{Ann}\mathcal{F})_{\text{red}}$ or of $\text{Ann}\mathcal{F}$ as they are homeomorphic. Notice that here we mean that each irreducible component of the support has dimension at most n .

The by the support being proper. We know that both $(\text{Ann}\mathcal{F})_{\text{red}}$ and $\text{Ann}\mathcal{F}$ are k -schemes, and notice that we have:

$$\begin{array}{ccc} (\text{Ann}\mathcal{F})_{\text{red}} & \hookrightarrow & \text{Ann}\mathcal{F} \\ & \searrow & \swarrow \\ & \text{Spec}k & \end{array}$$

The standard properties of proper morphisms tells you that $(\text{Ann}\mathcal{F})_{\text{red}}$ is proper over k iff $\text{Ann}\mathcal{F}$ is proper over k , therefore it doesn't matter which one we need to be proper.

Now we are able to explain why we need the assumption of proper of dimension n . Recall that by assumption we have:

$$\text{Ann}\mathcal{F} \cdot \mathcal{F} = 0$$

Recall that this is defined to be the sheafification of the presheaf where on each U we have:

$$\text{Ann}(U)\mathcal{F}(U) = 0$$

and the sheafification of 0 is also 0. Recall that we have the following general result on closed subschemes. If $Y \hookrightarrow X$ is a closed subscheme with the exact sequence:

$$0 \rightarrow \mathcal{I} \rightarrow \mathcal{O}_X \rightarrow i_*\mathcal{O}_Y \rightarrow 0$$

such that a quasicoherent sheaf $\mathcal{G}$ on X is annihilated by $\mathcal{I}$ in the sense that $\mathcal{I}\mathcal{G} = 0$. Then:

1. $i^*\mathcal{G}$ is canonically isomorphic to $i^{-1}\mathcal{G}$ as a quasicoherent $\mathcal{O}_Y$ -sheaf where $i^{-1}\mathcal{G}$ carries a canonical quasicoherent $\mathcal{O}_Y$ -sheaf structure.
2. We have a canonical isomorphism of quasicoherent $\mathcal{O}_X$ sheaves:

$$\mathcal{G} \rightarrow i_*i^{-1}\mathcal{G}$$

Recall that the pushforward of a quasicoherent sheaf by a closed embedding is also quasicoherent, therefore this makes sense. I won't write out all the details, but the key is to notice that i^{-1} is exact and the following canonical isomorphisms:

$$i^*\mathcal{G} = i^{-1}\mathcal{G} \otimes_{i^{-1}\mathcal{O}_X} \mathcal{O}_Y, \quad \mathcal{O}_Y \simeq i^{-1}\mathcal{O}_X / i^{-1}\mathcal{I}$$

Then know these facts, recall that when we compute the Čech cohomology, we have canonical isomorphisms:

$$H^i(Y, \mathcal{G}) \simeq H^i(X, i_*\mathcal{G})$$

Moreover, the Euler Characteristics χ is defined by alternating sums of cohomology, so we have:

$$\chi(X, \mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F}) = \sum_{i=0}^{\dim X} (-1)^i \dim_k H_i(X, \mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F})$$

Here the dimension of X , even if X is not irreducible and has multiple irreducible components with different dimension, $\dim X$ is the largest dimension of all its components.

Then there is the question that is this a well defined integer which means we need to make sure that each of the cohomology group is finite dimensional. If X is proper or projective itself, then we are good because Grothendieck's coherent theorem tells you that each of the dimension is finite. However, if X is not proper itself. Then we can use the k -vector space isomorphism:

$$H_i(X, \mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F}) \simeq H_i(\text{Supp}\mathcal{F}, i_*i^{-1}\mathcal{L}_{i_1}^\vee \otimes \cdots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F})$$

Notice that since $\mathcal{F}$ is annihilated by $\text{Ann}\mathcal{F}$, the tensor product also gets annihilated by $\text{Ann}\mathcal{F}$. Then if $\text{Supp}\mathcal{F}$ is proper, we can now apply Grothendieck's Coherence Theorem.

Finally, always notice that whenever we write $(\mathcal{L}_1^\vee, \dots, \mathcal{L}_n^\vee, \mathcal{F})$, we are always implying that the dimension of $\mathcal{F}$ is at most n .

Remark 3.1. Recall that if Y is a closed subscheme of X via the closed embedding:

$$i : Y \hookrightarrow X$$

with dimension at most n , we will also write:

$$(\mathcal{L}_1, \dots, \mathcal{L}_n, Y)$$

which just simply means that:

$$(\mathcal{L}_1, \dots, \mathcal{L}_n, i_* \mathcal{O}_Y)$$

Indeed, the pushforward of the structure sheaf by a closed embedding is still coherent. Moreover, if you compute the support of $i_* \mathcal{O}_Y$, you get Y , which means that the support of $i_* \mathcal{O}_Y$ is indeed again at most n .

Another way you can think about this is via the projection formula **** review*, since the tensor product of line bundles still remain, moreover we have natural isomorphisms for all line bundles $\mathcal{L}$ on X that:

$$i_* \mathcal{O}_Y \otimes \mathcal{L} \simeq i_*(i^* \mathcal{L} \otimes \mathcal{O}_Y)$$

Therefore if we are computing the cohomology of $\mathcal{L}$ we always get:

$$H^i(X, \mathcal{L} \otimes i_* \mathcal{O}_Y) = H^i(Y, i^* \mathcal{L} \otimes \mathcal{O}_Y) = H^i(Y, i^* \mathcal{O}_Y)$$

Sometimes this way to think of everything happening on Y may be more useful.

Obviously, if we are writing:

$$(\mathcal{L}_1, \dots, \mathcal{L}_n)$$

We just mean:

$$(\mathcal{L}_1, \dots, \mathcal{L}_n, \mathcal{O}_X)$$

are we are leaving $\mathcal{O}_X$ implicit.

Remark 3.2. To give some geometric intuition about why this is called the intersection product. Recall that for each Cartier divisor on any scheme X , we can consider the associated line bundle:

$$\mathcal{O}_X(D)$$

We also know that the ideal defining D is the dual:

$$\mathcal{O}_X(-D)$$

Therefore for the intersection product, if we are intersecting something with the line bundle associated to D , what we are really thinking of geometrically is that we are intersecting the support of that coherent sheaf with D which is cut out by the ideals, which is why we take the dual in the definition.

For example, if we consider the 0 divisor, which means that its line bundle is $\mathcal{O}_X$, then intersecting with the 0-divisor doesn't contribute anything, which should give us:

$$(\mathcal{O}_X, \mathcal{L}_2, \dots, \mathcal{L}_n, \mathcal{F}) = 0$$

Indeed, this is the following proposition:

Proposition 3.3.

$$(\mathcal{O}_X, \mathcal{L}_2, \dots, \mathcal{L}_n, \mathcal{F}) = 0$$

Proof. Don't forget the empty subset of $\{1, \dots, n\}$ and you can easily check it. □

We should also say a few words on why we require the dimension of the support of $\mathcal{F}$ to be at most n when we are writing $(\mathcal{L}_1, \dots, \mathcal{L}_n, \mathcal{F})$. Notice that since we already require the support of $\mathcal{F}$ to be proper, according to the previous definitions, we actually know that even if we don't assume the dimension of $\text{Supp} \mathcal{F} \leq n$, the intersection number can still be defined. However, geometrically speaking, this doesn't look like an intersection number in the classical sense. Concretely speaking, back to our geometric picture that each of the line bundles here should represent a divisor D . Then if we are intersecting something dimension n with n divisors, we expect to get something dimensions 0, which are just a set of points, possibly with multiplicities. However, if we start with something with dimension $n + 1$, intersecting with n divisors could give you something dimension 1 which doesn't make sense in terms of what the intersection number is trying to capture.

Proposition 3.4. Show that the intersection number:

$$(\mathcal{L}_1, \dots, \mathcal{L}_n, \mathcal{F})$$

on X is equal to:

$$((\mathcal{L}_1) \otimes_k K, \dots, (\mathcal{L}_n) \otimes_k K, \mathcal{F} \otimes_k K)$$

where K/k is any field extension of k .

Proof. Let's first recall that for locally free finite rank sheaves, pullback commutes with taking duals. ***
review that pullback commutes with tensors and everything works locally

*** *review that there is a canonical isomorphism from inverse image of tensors to tensors of inverse images, which gives the canonical isomorphism between pullback of tensors to tensors of pullback.*

Therefore, if we use $p : X_K \rightarrow X$ to denote the canonical projection, we know that $\mathcal{L}_i \otimes_k K$ is just the pullback of $\mathcal{L}_i$ along p , we know that we have equality:

$$\chi(p^*(\mathcal{L}_{i_1})^\vee \otimes \dots \otimes p^*(\mathcal{L}_{i_m})^\vee \otimes p^*\mathcal{F}) = \chi(p^*(\mathcal{L}_{i_1}^\vee \otimes \dots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F}))$$

Recall that the Euler characteristic is defined by the dimension of $H^i(X_K, p^*(\mathcal{L}_{i_1}^\vee \otimes \dots \otimes \mathcal{L}_{i_m}^\vee \otimes \mathcal{F}))$, but we know that pullback of the Cech cohomology doesn't change dimension. Therefore we get the same Euler characteristic. $\square$

3.2 Chern Classes

3.3 Algebraic Hartog's lemma

In this section we mainly prove the following result:

Theorem 3.5. (Algebraic Hartog's lemma) Let A be an integrally closed Noetherian integral domain, then we have:

$$A = \bigcap_{\text{codim}=1} A_{\mathfrak{p}}$$

Here this intersection is taken in $K(A)$.

To prove this result, there are a few algebraic preliminaries. First, a Noetherian dimension 1 local normal ring is a DVR and a DVR is regular local of dimension 1. It's interesting to see how a dimension 1 Noetherian local normal ring R is a DVR. Recall that to prove it's a DVR, we need to show that R is a PID. Let's first pick some element $x \in \mathfrak{m}$ where $\mathfrak{m}$ is the unique maximal ideal. Our intuition is that (x) should be some power of $\mathfrak{m}$ if R is a DVR, therefore we want to take the minimal n such that:

$$\mathfrak{m}^n \subset (x)$$

Or in other words, $\mathfrak{m}^{n-1} \not\subset (x)$.

You obtain this from the fact that $\sqrt{(x)} = \mathfrak{m}$, then, from the fact that R is Noetherian, we know that there is actually some n such that $\mathfrak{m}^n \subset (x)$. Then we can just choose the smallest possible n .

This implies that there is a:

$$y \in \mathfrak{m}^{n-1}, \quad y \notin (x)$$

We are thinking of this y as some sort of $u\pi^{n-1}$ where u is a unit. This implies that $\frac{y}{x} \notin R$, and say we denote this quotient by z . In fact we are thinking of this z as some $v\pi^{-1}$. Notice that we have $y\mathfrak{m} \subset \mathfrak{m}^n \subset (x)$ which implies that $z\mathfrak{m} \subset R$. However, I claim that $z\mathfrak{m} \not\subset \mathfrak{m}$. Indeed, this should happen as our intuition from DVR is that $z\mathfrak{m}$ should be the whole ring R . The way we show it in a Noetherian normal ring is by the following procedure, if $z\mathfrak{m} \subset \mathfrak{m}$, then since R is Noetherian, we have some matrix representation of $\mathfrak{m} \xrightarrow{\times a} \mathfrak{m}$. Then the usual determinant trick gives you some integral equation of z in terms of determinants. This implies $z \in R$, which is a contradiction. But this already enough since we know that there is some $a \in \mathfrak{m}$ such that za is a unit u in R , which means that $au^{-1} = \frac{1}{z}$ is now in R . We denote this to be π and we see that for all $m \in \mathfrak{m}$ we have:

$$zm \in R \Rightarrow m = \pi r \in (\pi)$$

Then let's prove the Algebraic Hartog's lemma and the ket construction is the generalization of the following observation:

Let A be an integral domain and let $r \in K(A)$, then we find that $r \in A$ iff $r \in A_{\mathfrak{p}}$ for all prime ideals $\mathfrak{p}$. This coincides with our usual notion of regular languages in complex manifolds, which says that a function regular everywhere should be a global regular function.

To prove this, if $r \in A$ then clearly it's in every $A_{\mathfrak{p}}$. The converse is the more interesting part, if $\frac{a}{b} \in A_{\mathfrak{p}}$, notice that even though we would like to conclude that b is not in $\mathfrak{p}$, this is actually not correct. Indeed, we can multiply both a, b by something in $\mathfrak{p}$ and still get r . To solve this ambiguity, recall that we can define for each $r \in K(A)$ the ideal:

$$I_r = \{a \in A : ar \in A\}$$

which is understood to be the ideal of the denominators of r . Then the points in $V(I_r)$ are exactly those points where r is not "regular" or in other words $r \notin A_{\mathfrak{p}}$. Indeed, we know that $r \in A_{\mathfrak{p}}$ iff there is some $a \notin \mathfrak{p}$ such that $ar \in A$ iff $\mathfrak{p} \notin V(I_r)$. Then we know that if $r \in A_{\mathfrak{p}}$ for all $\mathfrak{p}$, then $V(I_r)$ is empty, which means $I_r = A$ which means that $1 \cdot r = r \in A$. The converse also holds.

The following construction actually gives you a more refined compared to I_r .

Suppose that $r = \frac{a}{b} \notin A$ but $a, b \in A$, then let $\bar{a}$ be the image of a in $A/(b)$. Notice that by our assumption $\bar{a} \neq 0$. Let $M^{\frac{a}{b}}$ be the A -submodule $A/(b)$ generated by $\bar{a}$. Then we find that:

$$\frac{a}{b} \in A_{\mathfrak{q}} \Leftrightarrow M_{\mathfrak{q}}^{\frac{a}{b}} = 0$$

Indeed, if you think about what is $M_{\mathfrak{q}}^{\frac{a}{b}}$ it's just $A_{\mathfrak{q}}/(b)$. Then clearly we know that if $\frac{a}{b} \in A_{\mathfrak{q}}$, then a is a multiple of $b \in A_{\mathfrak{q}}$, therefore $(\bar{a})$ should be 0 in $A_{\mathfrak{q}}/(b)$. Conversely, if $(\bar{a}) = 0$ in $A_{\mathfrak{q}}/(b)$, then we again get the conclusion $\frac{a}{b} \in A_{\mathfrak{q}}$.

Lemma 3.6. If A is a Noetherian integral domain and let $r \in K(A)$, then $r \in A$ iff for every prime $\mathfrak{p}$ in $\text{Ass}(A/(g))$ for every nonzero $g \in A$, we have $r \in A_{\mathfrak{p}}$.

Proof. Notice that if $r \in A$, then clearly $r \in A_{\mathfrak{p}}$ for all $\mathfrak{p}$. To prove the converse, let's assume $r = \frac{a}{b} \notin A$, then our job is to find some g and one of it's associated prime $\mathfrak{q}$ such that $r \notin A_{\mathfrak{q}}$. Consider $M^{a/b} \subset A/(b)$. Then since $r \notin A$, we know that $(\bar{a})$ is not 0 in $A/(b)$, which implies that it has an associated prime $\mathfrak{p}$, which satisfies $M_{\mathfrak{p}}^{\frac{a}{b}} \neq 0$ which means that $\frac{a}{b} \notin A_{\mathfrak{p}}$. Since $(\bar{a})$ is a submodule of $A/(b)$, we know that $\mathfrak{q}$ is also an associated prime of $A/(b)$. $\square$

Remark 3.7. The above lemma has already gave us something about how to write the ring in terms of some intersections. That is if we only consider the primes which is an associated prime of A/g for some nonzero $g \in A$. We already have:

$$A = \bigcap_{\mathfrak{p}} A_{\mathfrak{p}}$$

Then to prove Algebraic Hartog's lemma, we have to somehow show that these primes are all codimension 1 primes (not saying that every codimension 1 prime should be an associated prime of A/g).

Proposition 3.8. If A is an integrally closed Noetherian domain then:

1. A is regular in codimension 1.
2. For any $b \in A$, the A -module $A/(b)$ has no embedded points.

Proof. We already proved 1 above since integrally closed is preserved by localization.

2 in the interesting part. Recall that since A is a Noetherian integral domain, then we know that the minimal primes of A/b are codimension 1, therefore we are just trying to show that for all b , the associated points of $A/(b)$ are codimension 1.

Now let $\mathfrak{p}$ be an associated point of $A/(b)$, we will prove that $\mathfrak{p}$ is of codimension 1 in $A_{\mathfrak{p}}$. Therefore it suffice to prove the following: If $(A, \mathfrak{m})$ is a Noetherian integral domain which is local and integrally closed, then for any $b \in \mathfrak{m} \setminus \{0\}$, suppose that $\mathfrak{m} \in \text{Ass}(A/(b))$, then $\mathfrak{m}$ is principal. This implies that $\mathfrak{m}$ is of codimension at most 1 and thus exactly 1 since (0) is also a prime ideal.

Now let's prove this statement. Choose $\bar{a} \in A/b$ such that $\text{Ann} \bar{a} = \mathfrak{m}$. Then first $\bar{a} \neq 0$, then we can lift it back to A and say get a . Then we know that $a/b \in K(A) \setminus A$. We also know that:

$$(a/b)\mathfrak{m} \subset A$$

I claim that $(a/b)\mathfrak{m} \not\subset \mathfrak{m}$, otherwise we know that again we use the determinant trick to conclude a/b is integral over A , which makes $a/b \in A$ since A is integrally closed. Then $(a/b)\mathfrak{m}$ is an ideal that contains a unit of A , thus is A . This implies:

$$\mathfrak{m} = \left(\frac{b}{a}\right)$$

Recall that again since A is integrally closed then $\frac{a}{b} \notin A$ implies $\frac{b}{a} \in A$, we are done. $\square$

Theorem 3.9. Let A be a Noetherian integral domain which is normal at codimension 1. Then the following two conditions are equivalent:

1. A is integrally closed.
2. $A = \bigcap_{\text{codim} \mathfrak{p}=1} A_{\mathfrak{p}}$.

Proof. Notice that if A is integrally closed, then the above proposition shows that for all $b \in A$:

$$A/(b)$$

has not embedded associated points. This and the above lemma implies that to see if $\frac{a}{b} \in A$ or not, we only need to consider associated primes of A/b and those are all codimension 1, therefore we only need to consider if they are in all $A_{\mathfrak{q}}$ for codimension 1 primes $\mathfrak{q}$.

Then let's assume $A = \bigcap_{\text{codim} \mathfrak{p}=1} A_{\mathfrak{p}}$. Let's assume that $r \in K(A)$ is integral over A , then it's integral over all $A_{\mathfrak{q}}$ for codimension 1 $\mathfrak{q}$. But recall that regular dimension 1 Noetherian local rings are integrally closed since this implies that it's maximal ideal is principal, thus a DVR (Have to use the fact that $\bigcap_i \mathfrak{m}^i = 0$ in Noetherian local rings due to Artin Rees lemma) thus integrally closed. But this implies that $r \in A_{\mathfrak{p}}$ for all codimension 1 $\mathfrak{p}$, thus by assumption $r \in A$. $\square$

Corollary 3.10. Suppose that A is a Noetherian integral domain satisfying conditions:

1. A is regular in codimension 1.
2. For any $b \in A$, the A -module $A/(b)$ has no embedded points.

Then it's integrally closed.

Proof. Notice that with condition 1 and 2, we can show that associated primes of A/g are all codimension 1 primes (Indeed the only assumption we need is it's regular at codimension 1 to show A/b has no embedded primes), therefore we know that we have:

$$A = \bigcap_{\text{codim} \mathfrak{p}=1} A_{\mathfrak{p}}$$

Then we use the above theorem. $\square$

Recall that on a locally Noetherian scheme X , we define **rational functions** on X to be an element in the image of the following map:

$$\Gamma(U, \mathcal{O}_U) \rightarrow \prod_{p \in \text{Ass}(X)} \mathcal{O}_{X,p}$$

where U is an open subset of X that contains all the associated points of X . And you may have noticed that if $U \subset V$ are both such opens containing all associated points, then $f \in \Gamma(V, \mathcal{O}_V)$ and its restriction in $\Gamma(U, \mathcal{O}_U)$ are mapped to the same element in the product.

To give some geometric intuition on this definition, suppose that we consider the following affine Noetherian scheme:

$$X = \text{Spec} k[x, y]/(xy)$$

We know that it is not reduced and has 2-associated points which are (x) and (y) . Notice that if we consider the distinguished open $D(x)$ and the functions on $D(x)$, notice that we only have $k[x]_x$. The function y which is not 0 on the original space are killed. Compare this to, for example $D(x)$ on $\text{Spec} k[x, y]$, where we have $k[x, y]_x$ and an injective map $k[x, y] \rightarrow k[x, y]_x$, therefore by passing to a smaller open, we actually enlarged the space of those "legal" functions and no original function is killed. SO we allow some sort of pole like $D(x)$ in $\text{Spec} k[x, y]$ but not in $k[x, y]/(xy)$.

Using this intuition, we see that if all $p \in \text{Ass}(X)$ are in $D(s)$, then we know that s is not in any of the associated points, which implies that inverting s will not kill any of the original functions. This is exactly the reason why we want the open to include all the associated points as this definition is trying to capture the notion of a generically defined function and by this notion we don't want to exclude those original regular functions.

Then we should say that since those functions are generically defined, then if they agree on a smaller open, which still contains all associated points, then two functions should be equal. This also matches our discussion above.

Finally, you should be able to see that our above definition is equivalent to:

$$\text{colim} \Gamma(U, \mathcal{O}_U), \quad U \supset \text{Ass}(X)$$

Corollary 3.11. If f is a nonzero rational function on a locally Noetherian normal scheme, then if f has no poles, then f is regular.

Proof. First recall that locally Noetherian normal schemes actually splits into a disjoint union of irreducible components X_i each of which is also open.

Moreover we know that normal schemes are also reduced, therefore each of such an irreducible component is in fact integral with a unique generic point η_i . Let's use $K(\eta_i)$ to denote the function field of each irreducible component. Then we know that in this case, our definition of rational functions can be just identified with:

$$\prod_i K(\eta_i)$$

In fact, we can do this identification since we can take disjoint opens around each η_i . I will let you fill in the details.

Then we know that if we have a rational function:

$$(f_i), \quad f_i \in K(\eta_i)$$

and p is a codimension 1 point, which is regular. We first know that it's contained in some irreducible components X_i , then we consider f_i , and we know that $f_i \in K(\eta_i)$, which means that we can ask if f_i is in $\mathcal{O}_{X,p} = \mathcal{O}_{X_i,p} \subset K(\eta_i)$.

By our assumption on f_i doesn't have a pole at p , we know that $f_i \in \mathcal{O}_{X,p}$. This discussion holds for every codimension 1 points. This actually means that if f_i is in $K(\eta_i)$, then for all the codimension 1 points p on X_i , f_i is in $\mathcal{O}_{X_i,p}$. I claim this implies that $f_i \in \Gamma(X_i, \mathcal{O}_{X_i})$.

To show this we can only work with an irreducible normal locally Noetherian scheme X . We know that for any $\text{Spec} A$ affine open in X , using Algebraic Hartog's lemma, we know that f_i is actually in A . Then

if we pick an affine open cover of X , we conclude that f_i is a regular function on each of the affine opens, which coincide on overlaps, thus coming from a global regular function, thus we are done. $\square$

Now suppose that we are given a locally free sheaf $\mathcal{E}$ on a locally Noetherian scheme, we can also try to define regular sections and rational sections on X . In fact, as usual, a regular section is just some section $s \in \Gamma(X, \mathcal{E})$.

Recall that, for any open $U \subset X$ which contains all the associated points of X , we still have injective maps:

$$\Gamma(U, \mathcal{E}|_U) \rightarrow \prod_{p \in \text{Ass} X} \mathcal{E}_p$$

Again, you check this on affine opens where $\mathcal{E}$ is trivial where injectivity basically follows from the case of the structure sheaf on the affine space. Then you cover U by such affine opens. Again, we can prove that a rational section is just the colimit of:

$$\text{colim}_{U \supset \text{Ass} X} \Gamma(U, \mathcal{E})$$

3.4 Line Bundles and Weil Divisors

In this section, let's only work with Noetherian normal schemes, which means that we know that we have only finitely many irreducible components, each of which are open and disjoint, thus a connected component.

I will refer the readers to [Vak25] for Basic definitions. We will only include what is important for our uses and interesting here.

Suppose that $\mathcal{L}$ is an invertible sheaf on X and s is a rational section not vanishing everywhere on any irreducible component of X (there are only finitely many). Recall that since X is normal, we can only think of the rational sections as a section defined in some U where U contains all the generic points of components of X . Notice that in this case, requiring a section not vanishing everywhere on a component is exactly to say that it's image in $\mathcal{E}_{\eta_i}$ is not 0 where η_i is the generic point of a component. We will say more about this nonvanishing condition later.

We can make sense of the valuation of a rational function s along an irreducible Weil divisor $[Y]$. Suppose that p is a codimension 1 point, then we can pick an affine open of that point p where $\mathcal{L}$ is trivial. Notice that this affine open necessarily contain the generic point of the component, therefore we can think of s as a rational function and we can think about the valuation of s at that point p . Notice that this valuation doesn't depend on the trivialization we choose since any of the two trivializations differ by an invertible function on the intersection, which has valuation 0.

Recall that if X is an integral Noetherian scheme and f is a rational function on X , which is just some function in $K(X)$, then there are only finitely many zeros and poles of f on X . Indeed, we can cover X by finitely many affine opens, thus we can just assume that X is affine $\text{Spec} A$. Then suppose that $f = \frac{a}{b}$ for $a, b \in A$. Then I claim that there are only finitely many codimension 1 primes such that the valuation of f is positive and finitely many such that the valuation of f is negative.

Indeed, we notice that we only need to show that for each $a \in A$, it only has finite many zeros. Notice that a has a zero at $\mathfrak{p}$ iff $a \in \mathfrak{p}$. These correspond to minimal primes in $A/(a)$ and since A is Noetherian, there are only finitely many.

Then we will denote the valuation of s along Y by $\text{val}_Y(s)$. This gives us the following definition:

$$\text{div}(s) = \sum_Y \text{val}_Y(s)[Y]$$

which is the Weil divisor determined by this rational function s . It's all called the **divisor of zeros and poles** of s .

Remark 3.12. Let's say something about this nonvanishing conditions. This condition is not some mysterious condition without which the definition we gave above doesn't behave well. This is simply a condition without which we can not even make that definition.

To see why, if s is a rational function that vanishes completely on the component X_i , then we know that if p is a codimension 1 point on X_i if you compute the valuation of s at p , you simply get ∞ . Clearly, we don't want something funny like $\infty[Y]$ to show up in the definition of a divisor.

Then there will be the following important group, called the **group of line bundles with rational sections**.

Note first that we have the set $\{(\mathcal{L}, s)\}$ which consists of pairs of line bundles and a rational section of the line bundle, which doesn't vanish completely on any component of X . Now if we have $(\mathcal{L}, s)$ and $\{(\mathcal{L}', s')\}$ and an isomorphism:

$$\varphi : \mathcal{L} \rightarrow \mathcal{L}'$$

such that $\varphi(s) = s'$ up to the equivalent of rational functions, then we want to view the two pairs as the same since as you will see, they doesn't make a difference in terms of determining a divisor. Therefore we will mod out this equivalence relation on the set $\{(\mathcal{L}, s)\}$. Then it actually is still a group under tensor products.

Let's be more careful in checking this. First we need to check that the product of the two section still doesn't vanish completely on any of the components. Recall that on each stalk of the generic points, the section in the tensor product turns to products in the ring and we know that product of nonzero elements is still nonzero. Then if $(\mathcal{L}, s')$ and $(\mathcal{L}', s')$ are two two line bundles with two rational section, note that s, s' actually correspond to a rational section on $\mathcal{L} \otimes \mathcal{L}'$ by considering the image of $s \otimes s'$ from the presheaf of tensors to the sheaf of tensors.

The identity is obvious $(\mathcal{O}_X, 1)$. To figure out what is the inverse of say $(\mathcal{L}, s)$, first we notice that since we require s to be nonvanishing on any component, we can actually choose an open U containing all the generic points where s is nowhere vanishing, the advantage of this is that we have an isomorphism of $\mathcal{O}_U$ -modules:

$$\mathcal{O}_U \rightarrow \mathcal{L}|_U, \quad 1 \mapsto s$$

Then we know that in this case, to give an $\mathcal{O}_U$ -module morphism from $\mathcal{L}|_U$ to $\mathcal{O}_U$, it suffice to specify where this rational section s goes, i.e. via the following diagram:

$$\begin{array}{ccc} \mathcal{O}_U & \xrightarrow{1 \mapsto s} & \mathcal{L}|_U \\ & \searrow 1 \mapsto f & \swarrow s \mapsto f \\ & \mathcal{O}_U & \end{array}$$

Then we know that if we consider the evaluation morphism:

$$\mathcal{L} \otimes \mathcal{L}^\vee \rightarrow \mathcal{O}_X$$

Notice that on U , if we consider the morphism φ defined from $\mathcal{L}|_U$ to $\mathcal{O}_U$ where you map s to 1, then we know that $s \otimes \varphi$ is mapped to $\varphi(s) = 1$ on $\mathcal{O}_U$. This is illustrated in the following diagram:

$$\begin{array}{ccc} (\mathcal{L} \otimes \mathcal{L}^\vee)^{\text{pre}} & & \\ \downarrow & \searrow & \\ \mathcal{L} \otimes \mathcal{L}^\vee & \xrightarrow{\text{ev}} & \mathcal{O}_X \end{array}$$

We know that the section we defined above is actually on the presheaf, which is mapped to 1 on U . But this implies it's corresponding section in the sheafified sheaf should also map to 1.

We are done.

Remark 3.13. It's important to notice that if t is an invertible function on $\mathcal{O}_X$, then we can consider the following morphism:

$$(\mathcal{L}, s) \xrightarrow{\times t} (\mathcal{L}, st)$$

Since t is an invertible function, then if you see that is the induced morphism on each of the stalks of $\mathcal{L}$, we obtain an isomorphism, therefore we know that this is an isomorphism. Therefore we have:

$$(\mathcal{L}, s) \sim (\mathcal{L}, st)$$

One special case of this is that if:

$$(\mathcal{L}, u), \quad (\mathcal{L}, v)$$

are two pairs where u, v are two rational sections. Notice that, by shrinking to an open where u, v are nowhere vanishing, we can again consider the following morphism:

$$\mathcal{O}_U \rightarrow \mathcal{L}|_U, \quad 1 \mapsto u$$

Since this is an isomorphism we know that some function $f \in \mathcal{O}_U(U)$ is mapped to v . We will use the symbol $\frac{v}{u}$ to denote the rational function. Even though it's not legal to do division in the line bundle, we are still using this division notation to indicate that how we obtained f .

Then we know that if we consider the canonical isomorphism:

$$\mathcal{O}_X \otimes \mathcal{L} \rightarrow \mathcal{L}$$

We notice that on U , the section of $\mathcal{O}_U \otimes \mathcal{L}|_U$ corresponding to $f \otimes v$ is mapped to u , this implies that:

$$(\mathcal{O}_X, \frac{u}{v}) \cdot (\mathcal{L}, v) = (\mathcal{L}, u)$$

In other words, we know that we can easily find the following formula:

$$\frac{(\mathcal{L}, u)}{(\mathcal{L}, v)} = (\mathcal{O}_X, \frac{u}{v})$$

Remark 3.14. Up until this point, we still haven't used the assumption that X is normal in any essential way. Notice that all the discussion can be done if we only assume that X is a Noetherian scheme that regular in codimension 1 and reduced.

The only place we need to be a bit careful is if we don't assume the scheme is normal, then we don't have the fact that each irreducible component is also the connected component and thus is open. But this actually doesn't matter since for each generic point of the irreducible component, we can find an open neighborhood of the generic point which is disjoint with other components. Then we know that by working with these opens, we can pretend that we are working with several integral parts, each disjoint with another, thus back to the normal scheme case.

Another concern is that if now p is a codimension 1 point, then maybe it will be contained in two distinct irreducible components, which causes ambiguity to the valuation. But this actually doesn't happen if we assume X is regular in codimension 1. Indeed, if p is contained in 2 components, then in the local ring there will be two distinct minimal primes, which makes it not a domain. However, $\mathcal{O}_{X,p}$ is regular and local, which is a domain. Therefore the valuation still is well defined.

Note that the function div yields a group homomorphism:

$$\text{div} : \{(\mathcal{L}, s)\} / \sim \rightarrow \text{Weil} X$$

Indeed, if we have an isomorphism $(\mathcal{L}, s) \simeq (\mathcal{L}', s')$, then we know that locally on a common trivialization, s, s' only differs by an invertible function, which means that they will have the same valuation. This tells you that this is well defined. We also need to check this is group homomorphism, suppose that $(\mathcal{L}, s)$ and $(\mathcal{L}', s')$ are two line bundles with 2 rational sections. We know that the

Remark 3.15. Given any nontrivial line bundle $\mathcal{L}$, there is always such a rational section s , as long as we stick to our assumption that X is Noetherian reduced and regular in codimension 1. To see why **** to complete*

To give an example of what the construction above is, recall that on $\mathbb{P}_k^1 = \text{Proj} k[x_0, x_1]$, we have line bundles $\mathcal{O}(n)$ for every integer $n \in \mathbb{Z}$. For example the line bundle $\mathcal{O}(1)$ is defined to be the line bundle that is trivial on $D_+(x_0)$ and on $D_+(x_1)$ glued together on the intersection by the invertible function $\frac{x_1}{x_0}$ on $D_+(x_0) \cap D_+(x_1)$. In other words the transition function from U_0 to U_1 is defined by multiplication by $\frac{x_1}{x_0}$ and conversely the transition function from U_1 to U_0 is defined by multiplication of $\frac{x_0}{x_1}$. Then we know that if $a\frac{x_1}{x_0} + b$ on U_0 and $c\frac{x_0}{x_1} + d$ in U_1 want to glue to a global section. We know that since the restriction of $a\frac{x_1}{x_0} + b$ to the intersection is still itself and under the transition map it will stand for the function:

$$a + b\frac{x_1}{x_0}$$

Therefore we know that in order for the above two prescribed local sections to define a global section. We have to have $a = d, b = c$ which means that after we choose the section on $D_+(x_0)$, we actually don't have any freedom for the local section on $D_+(x_1)$. Therefore we can just think of the global sections of $\mathcal{O}(1)$ to be of the form $a\frac{x_1}{x_0} + b$. But this looks a lot ugly and not canonical in the sense that we can also first determine $c\frac{x_0}{x_1} + d$. Therefore to make global function look more canonical, we define it to be the 2-dimensional k -vector space spanned by the two symbols x_0, x_1 , which means that global sections now look like $ax_1 + bx_0$ and the restriction map to $D_+(x_0)$ will just be to formally divide by x_0 which means we have $a\frac{x_1}{x_0} + b$ and similarly the restriction to $D_+(x_1)$ will be formally divide by x_1 which gives us $a + b\frac{x_0}{x_1}$ clearly these two sections agree on the intersection via the transition map.

Then we look at:

Example 3.16. Let's try to make sense of what is the divisor determined by the rational section $\frac{x^2}{x+y}$ on $\mathbb{P}_k^1$.

First, notice that since $\mathbb{P}_k^1$ is integral of dimension 1 and is regular, a rational section is just a section defined on any open subset. The above rational section will be interpreted to be the restriction of the global section x to $D_+(x+y)$ times the function $\frac{x}{x+y}$ defined on $D_+(x+y)$. To compute the divisor, we pick the cover $D_+(x)$ and $D_+(y)$. We know that the rational section restricted to $D_+(x) \cap D_+(x+y)$ is:

$$1 \cdot \frac{x}{x+y}$$

Technically speaking we know that on $D_+(x) \cap D_+(x+y)$ the function $\frac{x}{x+y}$ can be thought as the inverse of $1 + \frac{y}{x}$ in $D_+(x)$. Then we know that the only divisor is the point:

$$(1 + \frac{y}{x})$$

which is a pole of order 1, which means that this corresponding prime in $k[x, y]$ is $(x+y)$. Then we consider its restriction in $D_+(x+y) \cap D_+(y)$ which is:

$$\frac{x}{y} \cdot \frac{x}{x+y}$$

Actually this notation is kind confusing since $\frac{x}{y}$ is given in terms in $D_+(y)$ but the function $\frac{x}{x+y}$ is in $D_+(x+y)$. To eliminate this confusion, we actually pass to the intersection, which can be thought of the 0 degree part of $k[x, y]_{y(x+y)}$, then the function $\frac{x}{x+y}$ can be thought of as the image of $\frac{\frac{x}{y}}{1 + \frac{x}{y}}$ under the identification. Then with the multiplication of the function $\frac{x}{y}$, we get a function:

$$\frac{(\frac{x}{y})^2}{1 + \frac{x}{y}}$$

This implies that on $D_+(y)$, we have two divisor points, where we have a zero of order 2 at $(\frac{x}{y})$, which correspond to the ideal (x) in $k[x, y]$. We have a pole of order 1 which still correspond to $(x+y)$ in $k[x, y]$. Combine this discussion, we know that the divisor is given by:

$$\text{div}(\frac{x^2}{x+y}) = 2[(x)] - [(x+y)]$$

Now we will examine the homomorphism div we define more carefully and it turns out that in a lot of contexts, this homomorphism completely determines all line bundles using Weil divisors.

Proposition 3.17. If X is normal and Noetherian then the map div is injective.

Remark 3.18. This is the proposition where we will specifically need the normality assumption.

Proof. Suppose that $\text{div}(\mathcal{L}, s) = 0$, then we know that in this case, in particular, this rational section has no poles. Since Noetherian normal schemes is just disjoint union of irreducible components. Let's check what happens on each irreducible component which is integral.

Notice that on each integral component, s has no poles, then by first passing to small affine opens where $\mathcal{L}$ trivializes, we can conclude that s is actually a regular section on this small affine by algebraic Hartog's lemma. Then we cover this component by these small affines and s is going to be a regular section on each of the affine. Notice that these regular sections agree on intersections since on integral schemes we know that the map $\mathcal{O}_{X,p} \rightarrow K(X)$ is injective and regular sections on each affine open correspond to the image of the rational section s in $K(X)$, therefore they agree on each stalk, thus agree. We can run this argument on each component of X and conclude that s is in fact a regular section of $\mathcal{L}$ (In other words, there is a unique regular section that restricts to s). Then we will just think of s as the global section.

Then I claim that s is nowhere vanishing, which implies that $\mathcal{O}_X \rightarrow \mathcal{L}, 1 \mapsto s$ is an isomorphism and we are done. In fact, we know s doesn't have any zeros on X , which implies that $s|_x$ for all $x \in X$ is not in the maximal ideal, which means that it doesn't vanish, therefore we are done. $\square$

Motivated by this result, we would like to know if div is surjective or at least in reasonable situations, try to determine its image. For the rest of this section, let's just assume X is **normal** and **Noetherian**. Let's assume that $X_1, \dots, X_n$ are the irreducible components and each with function field $K(X_i)$.

Suppose that D on X is a Weil divisor. Let's define a sheaf determined by D , denoted by $\mathcal{O}_X(D)$ by on each $U \subset X$, we assign the sections to be:

$$\Gamma(U, \mathcal{O}_X(D)) = \{t \in \prod_{i=1}^n K(X_i)^\times : X_i \cap U \neq \emptyset, \text{div}|_U t + D|_U \geq 0\}$$

Recall that here $\text{div}|_U$ is just to only compute the valuation of t for those codimension 1 points in U and the divisor at a regular codimension 1 point will be $\{p\} \cap U$. Here the restriction of D to U is again by taking intersections with U .

One way you can think about the sections of $\mathcal{O}_X(D)$ over U are just rational functions on U whose zeros and poles are constrained by the divisor $D|_U$. From the definition, you can see that a positive coefficient in D allows for a pole of the rational function t and conversely, a negative coefficient in D allows for a zero of the rational function t . Note that if U doesn't intersect D at all, we find that we are only allowed to take rational functions who don't have a pole, but then by Algebraic Hartog's lemma, we know that these functions have to be regular functions.

Remark 3.19. It's very **important** to note that if U doesn't intersect with D but still contains all the generic points, there will be a very important rational section of $\mathcal{O}_X(D)$, which is 1 in $K(X)^\times$.

It turns out that instead of directly working with the reducible case, let's first see what happens if X is irreducible which is helpful for us to understand the reducible case. Note that if X is irreducible normal Noetherian, then:

$$\Gamma(U, \mathcal{O}_X(D)) = \{t \in K(X)^\times : \text{div}|_U t + D|_U \geq 0\} \cup \{0\}$$

Note that this $\mathcal{O}_X(D)$ is always a quasicoherent sheaf and in particular it's actually a sheaf. Note that the identity axiom is clearly satisfied and if we have an open cover U_i of U on which U_i we have t_i which agrees on the intersections (in fact we only need the intersection of 2 opens), then clearly they are the same element t in $K(X)^\times$. I will let you see that t is in fact an element in $\mathcal{O}_X(U)$ as well. To show this is quasicoherent, note that if U is affine, say $\text{Spec} A$, then we can think of $\Gamma(U, \mathcal{O}_X(D))$ to be a subset of $K(A)^\times$. Say we restrict to a distinguished open $D(f)$, notice that on $D(f)$, the function f is invertible thus has valuation 0 at every codimension 1 point in $D(f)$. Therefore we know that whatever is in $\Gamma(U, \mathcal{O}_X(D))$, we can divide it by powers of f , and they will still be in $\Gamma(D(f), \mathcal{O}_X(D))$. Conversely, we know that if some

$t \in \Gamma(D(f), \mathcal{O}_X(D))$, since f is a regular function on $\text{Spec} A$ for codimension 1 points outside $D(f)$, f has at least a zero of degree 1 at each of the point, therefore we know that tf^n for some large n is in $\Gamma(U, \mathcal{O}_X(D))$. Therefore the restriction from $U \simeq \text{Spec} A$ to $V \simeq D(f)$ is isomorphic to localization at f , therefore it's quasicohherent.

Now if X is not irreducible, notice that the definition:

$$\Gamma(U, \mathcal{O}_X(D)) = \{t \in \prod_{i=1} K(X_i)^\times : X_i \cap U \neq \emptyset, \quad \text{div}|_U t + D|_U \geq 0\} \cup \{0\}$$

Now is just a product of $\Gamma(U_i, \mathcal{O}_{X_i}(D))$ where X_i are the irreducible components of X and $U_i = U \cap X_i$. Recall that this open embedding X_i to X is also at the same time a closed embedding. Therefore we may think of $\mathcal{O}_X(D)$ as the product of the pushforward of $\mathcal{O}_{X_i}(D \cap X_i)$. Clearly what happens on each component doesn't affect what happens on other components, therefore let's just stick to the assumption that X is integral because we can always work on each component respectively,

Example 3.20. Let's see an example after the above definition. Let $X = \mathbb{A}_k^1$ and consider the divisor:

$$D = -2[(x)] + [(x-1)] + [(x-2)]$$

We see that a global section of $\mathcal{O}_X(D)$ is just $\frac{3x^2}{x-1}$ since we know that this global section has 1 zero (x) and a pole at $(x-1)$ which is both allowed by D . This global section even has one more zero at (x) to spare and one more pole at $(x-2)$ to spare.

I claim that any global section of $\mathcal{O}_X(D)$ is going to be a $k[x]$ -multiple of $\frac{x^2}{(x-1)(x-2)}$. Suppose that $\frac{f(x)}{g(x)} \in k(x)$ is a rational function, then it's a global section of $\mathcal{O}_X(D)$ if it has at least two zeros at (x) and at most 1 pole at both $(x-1)$ and $(x-2)$ and no other poles at any other prime. Suppose that f, g already has no common irreducible factors, then if this is a global section, we know that our assumption implies that $g(x)$ the only two irreducible factors of g can be $(x-1)$ and $(x-2)$ and $f(x)$ at least has a factor of x^2 . This is clearly a $k[x]$ -multiple of $\frac{x^2}{(x-1)(x-2)}$.

Then consider the following morphism of quasicohherent $\mathcal{O}_X$ -modules:

$$\mathcal{O}_X \rightarrow \mathcal{O}_X(D), \quad 1 \mapsto \frac{x^2}{(x-1)(x-2)}$$

I claim that this is an isomorphism, which implies that $\mathcal{O}_X(D)$ is an invertible sheaf. Indeed, suppose that $g \in \mathcal{O}_X(D)(U)$, notice that this implies that:

$$\text{div}(g) + D|_U \geq 0$$

Notice that we can think of D as the divisor defined by $\frac{(x-1)(x-2)}{x^2}$, therefore we know that we are just computing:

$$\text{div}(g \frac{x^2}{(x-1)(x-2)})|_U \geq 0$$

This implies that $g \frac{(x-1)(x-2)}{x^2}$ is a regular section on U , therefore we know that g is the image of $g \frac{(x-1)(x-2)}{x^2} \in \mathcal{O}_X(U)$. This shows this is surjective even in the level of sections. This is also clearly injective since everything is defined in $k(x)$.

Example 3.21. Let's examine further about the previous example. For concreteness say $k = \mathbb{C}$. Since $\mathcal{O}_X(D)$, where D is the divisor given in the previous example, is a line bundle (in fact trivial), we can talk about the zeros and poles of a regular or rational section of $\mathcal{O}_X(D)$.

Consider a section of $\mathcal{O}_X(D)(X)$ corresponding $\frac{x^2(x-3)}{(x-1)(x-2)}$, then I claim that it only vanishes at 1 point. Indeed, we know that this point, under the isomorphism $\mathcal{O}_X \rightarrow \mathcal{O}_X(D)$ comes from the global section $(x-3)$. Therefore it only vanishes with order 1 at the point $(x-3)$.

Another way to see this is to think about the stalk of $\mathcal{O}_X(D)$. Notice that if $p = (f(x))$ is not (x) , $(x-1)$ and $(x-2)$, then clearly the stalk at p is just $k[x]_{(f(x))}$ and a $k[x]_{(f(x))}$ module, the way you restrict the global section $\frac{x^2(x-3)}{(x-1)(x-2)}$ to the stalk, it's still $\frac{x^2(x-3)}{(x-1)(x-2)}$. If p is (x) or $(x-1)$ or $(x-2)$, we know that for

example the stalk of $\mathcal{O}_X(D)$ at (x) is just $x^2k[x]_{(x)}$. The this stalk is just $\frac{1}{(x-1)}k[x]_{(x-1)} \subset k(x)$. Then I will let you fill in the rest of the details.

Then using the same argument, you see that the global section $x^2(x-1)$ vanishes at two places. It vanishes with order 2 at $(x-1)$ and vanishes with order 1 at $(x-2)$.

Finally, recall that we have a canonical rational section 1 for $\mathcal{O}_X(D)$. We can use the above argument to check out it's zeros and poles and you can find that it's zeros are at $(x-1)$ and $(x-2)$ both with order 1 and it's poles are only at (x) with an order of 2.

Next we will try to find a criterion when $\mathcal{O}_X(D)$ is an invertible sheaf. This will also give you the image of the map div in Weil.

Theorem 3.22. Suppose that X is a normal Noetherian integral scheme and let $\mathcal{L}$ be an invertible sheaf with a rational section s . Then:

1. There is a canonical isomorphism:

$$\mathcal{O}_X(\text{div}(s)) \rightarrow \mathcal{L}$$

2. Let σ be the map from $K(X)$ to the rational section of $\mathcal{L}$ defined by the above isomorphism. That is to say that for each rational section t with domain of definition U , it is mapped via the isomorphism to a rational section of $\mathcal{L}$. Then the canonical section 1 is mapped to s .

Assuming this theorem, we have the following diagram:

$$\{(\mathcal{L}, s)\} / \sim \begin{array}{c} \xrightarrow{\text{div}(s)} \\ \xleftarrow{(\mathcal{O}_X(D), 1)} \end{array} \{\text{Weil Divisors } D \text{ such that } \mathcal{O}_X(D) \text{ is a line bundle}\}$$

Then if we can show that for a Weil divisor D such that $\mathcal{O}_X(D)$ is a line bundle, then we know that $(\mathcal{O}_X(D), 1)$ is mapped by div to D , then we can identify the image of div with those Weil divisors such that the associated sheaf is invertible. Indeed, we have the following result:

Proposition 3.23. Let X be a Noetherian and normal integral scheme, suppose that D is a Weil divisor such that $\mathcal{O}_X(D)$ is a line bundle, then D is locally principal (which we will define later) and we have:

$$\text{div}((\mathcal{O}_X(D), 1)) = D$$

We will prove this later but for now, let's first prove the theorem above:

Proof. Let's first describe a morphism $\mathcal{O}_X(\text{div}(s)) \rightarrow \mathcal{L}$.

We will temporarily define two sheaves in this proof, which can be defined in more general settings we don't care. Let $\mathcal{K}_X$ be the sheaf on X such that for each $U \subset X$:

$$\mathcal{K}_X(U) = K(X)$$

with identity as the restriction. Clearly this is a sheaf since X is irreducible (if it's not irreducible, we will exactly meet the issue with the constant sheaf $\mathbb{Z}$). Notice that if we think about the stalk of $\mathcal{L}$ at the generic point η , denoted by $\mathcal{L}_\eta$, we can also define a sheaf $\mathcal{R}_X$ where on each U , we define the sections to be $\mathcal{L}_\eta$. Then clearly we have a natural inclusions:

$$\mathcal{O}_X(\text{div}(s)) \hookrightarrow \mathcal{K}_X, \quad \mathcal{L} \rightarrow \mathcal{R}_X$$

Then notice that we also have a canonical morphism of sheaves:

$$\mathcal{K}_X \rightarrow \mathcal{R}_X, \quad t \mapsto ts$$

where s is the rational section of $\mathcal{L}$. I claim that this induces an isomorphism of $\mathcal{O}_X(D) \rightarrow \mathcal{L}$, i.e. we have the following diagram:

$$\begin{array}{ccc} \mathcal{O}_X(\text{div}(s)) & \xrightarrow{\simeq} & \mathcal{L} \\ \downarrow & & \downarrow \\ \mathcal{K}_X & \xrightarrow{\times s} & \mathcal{R}_X \end{array}$$

To see why, first notice that on each U , if we consider the image of $t \in \mathcal{O}_X(\operatorname{div}(s))$ which is mapped to ts , as an element in $\mathcal{L}_\eta$. I claim that this element is also in $\mathcal{L}(U)$.

Indeed, by our definition of t we know that:

$$\operatorname{div}|_U(t) + \operatorname{div}(s)|_U \geq 0$$

Then suppose that we choose an affine open cover U_i of U where on each U_i , $\mathcal{L}$ trivializes and s correspond to a rational function f_i . We know that:

$$\operatorname{div}|_U(tf_i) \geq 0$$

Then we know that since tf_i has no pole on U_i , Algebraic Hartog's lemma tells us tf_i is a regular section on U_i , which means that ts is also a regular section in $\mathcal{L}(U_i)$. This holds for all U_i and on intersections, clearly they match, therefore we get a global section $ts \in \mathcal{L}(U)$. This shows that the morphism defined by multiplying s indeed induces a morphism from $\mathcal{O}_X(\operatorname{div}(s))$ to $\mathcal{L}$.

This is clearly injective since the morphism from $\mathcal{K}_X$ to $\mathcal{R}_X$ is injective. To show it's surjective, we only have to show it locally as surjectivity for sheaves are determined locally. I claim that the open subsets U on which $\mathcal{O}_X(\operatorname{div}(s))|_U \simeq \mathcal{O}_U$ forms a basis for the Zariski topology of X . Indeed, suppose that U is an affine open such that $\mathcal{L}$ trivializes, where s correspond to a rational function f on U , then consider the map:

$$\mathcal{O}_U \rightarrow \mathcal{O}_X(\operatorname{div}(s))|_U, \quad 1 \mapsto \frac{1}{f}$$

Indeed, we know that:

$$\operatorname{div}(s)|_U = \operatorname{div}(f)|_U \Rightarrow \frac{1}{f} \in \mathcal{O}_X(\operatorname{div}(s))(U)$$

This is an isomorphism since we know that for any $g \in \mathcal{O}_X(\operatorname{div}(s))(U)$, we know that:

$$\operatorname{div}|_U(gs)|_U = \operatorname{div}|_U(gf) \geq 0$$

This implies that gf is in $\mathcal{O}_X(U)$, therefore we know that $g = a \frac{1}{f}$ for some $a \in \mathcal{O}_X(U)$. Injectivity is clear. The affine opens where $\mathcal{L}$ trivializes forms a basis, which proves our claim.

Then on each of such affine open U , consider:

$$\mathcal{O}_U \rightarrow \mathcal{O}_X(\operatorname{div}(s))|_U \rightarrow \mathcal{L}|_U \rightarrow \mathcal{O}_U$$

where $1 \mapsto \frac{1}{f} \mapsto \frac{s}{f} \mapsto 1$. This is indeed an isomorphism which proves that $\mathcal{O}_X(\operatorname{div}(s))|_U \rightarrow \mathcal{L}|_U$ is an isomorphism, in particular surjective. This proves 1.

To prove 2, we need to figure out where does the canonical section 1 in $\mathcal{O}_X(\operatorname{div}(s))$ go to. Notice that the image of 1 in $\mathcal{R}_X$ is just s , which means that it can only come from s in $\mathcal{L}$, we are done. $\square$

Remark 3.24. Conceptually speaking, the above morphism from $\mathcal{O}(\operatorname{div}(s)) \rightarrow \mathcal{L}$ is just t maps to ts . However, we have some trouble defining this morphism directly as s is just a regular section defined on some open subset and t is just an element in $K(X)$, so we need to make sense of what is ts . The above proof is just gives you a formal framework where we make this morphism rigorous.

Example 3.25. Let $X = \mathbb{P}^n$, then we already know that this is a Noetherian normal integral scheme therefore we can apply the results we obtained before.

Let $\mathcal{L} = \mathcal{O}(1)$ and let $s = x_0$ be the global section. We will consider $D = \operatorname{div} x_0$. What we already know is that $\mathcal{O}(D) \simeq \mathcal{O}(1)$ by the above theorem. Now we will verify that $\mathcal{O}(mD) \simeq \mathcal{O}(m)$ for all $m \in \mathbb{Z}$.

Let's show that $\mathcal{O}(mD)$ is trivial on each $D_+(x_i)$ and the transition maps from $D_+(x_i)$ to $D_+(x_j)$ agrees with that of $\mathcal{O}(m)$ which is $\frac{x_i}{x_j}^{-m}$. Notice that $D = \operatorname{div} x_0$ and since x_0 only vanishes on one codimension 1 point, which is (x_0) . Then we know that for example on $U = D_+(x_0)$, $\mathcal{O}(mD)$ is just $\mathcal{O}_U$. For example on other $U = D_+(x_i)$, we know that $\mathcal{O}(mD)$ is $(\frac{x_0}{x_i})^{-m} \mathcal{O}_U$. (Do you see why these are true?) Clearly on each $D_+(x_i)$, $\mathcal{O}(mD)$ is trivial. Notice that on $U_i \cap U_j$, if we want to compute the transition function we have:

$$\left(\frac{x_0}{x_i}\right)^{-m} \mathcal{O}_{U_i} \xrightarrow{\times \left(\frac{x_i}{x_j}\right)^{-m}} \left(\frac{x_0}{x_j}\right)^{-m} \mathcal{O}_{U_j}$$

This is exactly the transition function of $\mathcal{O}(m)$. This gives us an isomorphism:

$$\mathcal{O}_X(mD) \rightarrow \mathcal{O}(m)$$

where on each $D_+(x_i)$ the morphism is defined to be $(\frac{x_0}{x_i})^{-m} \mathcal{O}_U \xrightarrow{\times (\frac{x_0}{x_i})^m} \mathcal{O}_U$.

Let's compute under this isomorphism where does the generic section 1 goes to. We can only work on $D_+(x_0)$ where the map is defined by:

$$\mathcal{O}_{D_+(x_0)} \xrightarrow{\text{Id}} \mathcal{O}_{D_+(x_0)}$$

Therefore we know that 1 is mapped to 1 on $D_+(x_0)$, which is the restriction of the global section x_0^m of $\mathcal{O}(m)$ to 1. Therefore the canonical section 1 is mapped to x_0^m .

Remark 3.26. Notice that the above discussion actually determines that $\text{Pic}\mathbb{P}_k^n = \mathbb{Z}$. Indeed we already see that $\mathbb{Z}$ is a subgroup of $\text{Pic}\mathbb{P}_k^n$. Now given any $\mathcal{L}$ and we consider the divisor determined by some rational section s . Say this divisor is:

$$a_1[Y_1] + \cdots + a_n[Y_n]$$

where Y_i is the irreducible closed subset defined by a homogeneous irreducible polynomial f_i with degree m_i . Notice that we know that $f_i \in \mathcal{O}(m_i)$ determines the divisor $[Y_i]$ which means that $\mathcal{O}(-m_i)$ and the corresponding section defines $-[Y_i]$. This is because div is injective.

Then if we consider $a_i[Y_i]$, we know that the section:

$$f_i \otimes \cdots \otimes f_i \in \mathcal{O}_X(m_i) \otimes \cdots \otimes \mathcal{O}(m_i) \simeq \mathcal{O}(a_i m_i)$$

defines the divisor $a_i[Y_i]$ via div . Then we know that for other Y_j we can use similar method and in conclusion, we know that the element:

$$f_1^{\otimes a_1} \otimes \cdots \otimes f_n^{\otimes a_n} \in \mathcal{O}(\sum_i a_i m_i)$$

defines:

$$a_1[Y_1] + \cdots + a_n[Y_n]$$

Therefore since div is injective $\mathcal{L} \simeq \mathcal{O}(\sum_i a_i m_i)$, therefore we know that the image of $\mathbb{Z} \hookrightarrow \text{Pic}\mathbb{P}_k^n$ is also surjective.

Finally we deal with the question of when a Weil divisor D makes $\mathcal{O}_X(D)$ a line bundle.

Definition 3.27. Let X be a Noetherian normal irreducible scheme and let D be a Weil divisor on X . If D is $\text{div} f$ for some rational function $f \in K(X)$, we say that D is **principal**. They clearly forms a subgroup of $\text{Weil}X$ and we will denote the subgroup by $\text{Prin}X$.

If X can be covered by open sets U_i such that $D|_{U_i}$ is principal on U_i , then we say that D is **locally principal**. This is also a subgroup of $\text{Weil}X$ and it's denoted by $\text{LocPrin}X$.

We have the following inclusion of groups:

$$\text{Prin}X \hookrightarrow \text{LocPrin}X \hookrightarrow \text{Weil}X$$

If D is principal, then $\mathcal{O}_X(D) \simeq \mathcal{O}$ and if D is locally principal, then $\mathcal{O}_X(D)$ is a line bundle. Indeed, if D is principal defined by $\text{div}(f)$, then we have a canonical isomorphism:

$$\mathcal{O}_X \rightarrow \mathcal{O}_X(D), \quad 1 \mapsto \frac{1}{f}$$

This is always injective and to show it's surjective we know that for any $g \in \mathcal{O}_X(D)$, we know that gf is a regular function on U , which means $g = \frac{a}{f}$. If D is locally principal, then we can define local trivialization morphisms, which shows it's a line bundle.

Conversely, we have:

Proposition 3.28. Let X be a Noetherian and normal integral scheme, suppose that D is a Weil divisor such that $\mathcal{O}_X(D)$ is a line bundle, then D is locally principal and we have:

$$\operatorname{div}((\mathcal{O}_X(D), 1)) = D$$

Proof. Pick a local trivialization of $\mathcal{O}_X(D)|_U$, say:

$$\mathcal{O}_U \rightarrow \mathcal{O}_X(D)|_U, \quad 1 \mapsto f$$

Then we know that every $g \in \mathcal{O}_X(D)(U)$ is a $\mathcal{O}_X(U)$ multiple of f . Then I claim that:

$$\operatorname{div} \frac{1}{f} = D|_U$$

Indeed, we know that since f is in $\mathcal{O}_X(D)(U)$:

$$\operatorname{div}|_U(f) + D|_U \geq 0$$

Suppose that $\operatorname{div}|_U(f) + D|_U > 0$. Suppose that those irreducible codimension 1 closed subsets with coefficients larger than 0 are $Y_1, \dots, Y_n$ with generic points $p_1, \dots, p_n$, say their coefficients are n_i . Let π_i be the uniformizer of the local rings $\mathcal{O}_{X,p_i}$, then we know that:

$$\operatorname{div}|_U\left(\frac{f}{\prod_i \pi_i^{n_i}}\right) + D|_U = 0$$

Therefore we know that by our assumption:

$$\frac{f}{\prod_i \pi_i^{n_i}} = fa \Rightarrow a = \frac{1}{\prod_i \pi_i^{n_i}}$$

We can assume that $U = \operatorname{Spec} A$ is integral affine, then we know that $a \in A$ iff $A \in A_{\mathfrak{p}}$ for all primes of A , but if we assume that p_i correspond to $\mathfrak{p}_i$ in $\operatorname{Spec} A$, $a \notin A_{\mathfrak{p}_i}$, which is a contradiction. Therefore:

$$\operatorname{div}|_U(f) = -D|_U$$

This implies that $D = \operatorname{div}|_U(\frac{1}{f})$.

Then consider what is the divisor determined by the canonical rational section 1. Notice that on each the above U , if we consider the isomorphism:

$$\mathcal{O}_U \rightarrow \mathcal{O}_X(D)|_U$$

defined above, we know that the rational function $\frac{1}{f}$ is mapped to the rational section 1, therefore we know that the divisor given by the rational section 1 is the same as the rational section $1/f$, which is $D|_U$, we are done $\square$

Finally, we can conclude that the following two maps are inverses of each other:

$$\{(\mathcal{L}, s)\} / \sim \xrightleftharpoons[(\mathcal{O}_X(D), 1)]{\operatorname{div}(s)} \{\text{Weil Divisors } D \text{ such that } \mathcal{O}_X(D) \text{ is a line bundle}\} = \{\text{Locally Principal Weil Divisors}\}$$

Recall that if a scheme X is Noetherian and factorial, then it's automatically normal, then we have:

Proposition 3.29. Let X be a Noetherian integral factorial scheme, then every Weil divisor D is locally principal, or in other words, every $\mathcal{O}_X(D)$ is a line bundle.

Proof. Let's first consider a simple case. Suppose that D is just $[Y]$ where Y is an irreducible closed codimension 1 subset.

We know that if we want to prove that D is locally principal, we need to find an open cover of X on each of the open $D \cap U$ is principal. First we consider the complement of D , and on the complement we know that D is cut out by 1. Then we consider an open cover of D by affine opens at each point y on Y . Notice that on each affine, say $\text{Spec} A$, Y correspond to a codimension 1 prime ideal $\mathfrak{q}$ and our point y correspond to a prime ideal $\mathfrak{p}$ with $\mathfrak{q} \supset \mathfrak{p}$. Notice that if we consider:

$$A_{\mathfrak{q}}$$

Then $\mathfrak{p}A_{\mathfrak{q}}$ is going to be a codimension 1 point and we know that it's going to be principal. Indeed, recall that in any UFD, any codimension 1 prime ideal is principal. This is because if we localize at that prime $\mathfrak{p}$, we get a UFD that is local of dimension 1, thus the prime is generated by some π coming from the UFD we started with which is also in this prime ideal. Then $(\pi) \subset \mathfrak{p}$. Notice that since π is irreducible in $A_{\mathfrak{p}}$, it's also irreducible in the original UFD, therefore (π) is also prime in A , therefore since $(\pi) \subset \mathfrak{p}$ and (π) is of codimension 1 by Krull's theorem, then we know that it has to be equal to $\mathfrak{p}$ since $\mathfrak{p}$ also has codimension 1.

Now we know in $A_{\mathfrak{q}}$ we have $(f) = \mathfrak{p}A_{\mathfrak{q}}$ where f is an element in $A_{\mathfrak{q}}$. We can assume that f actually is a function on a smaller affine open around $\mathfrak{q}$, say a distinguished open. Then we know that on that affine open if we consider (f) and $\mathfrak{p}$, we know that after we localize at $\mathfrak{q}$, they are equal to each other, then since everything is Noetherian, $\mathfrak{q}$ is finitely generated, therefore there is a further small affine open near $\mathfrak{q}$ such that $\mathfrak{p}$ is equal to (f) on that affine open. Notice that this is an affine open containing y and we know that $(f) = \mathfrak{p}$ on that open, therefore we know that if we consider the function f , it only has such a single zero at $\mathfrak{p}$. Since otherwise we know that $\mathfrak{p}$ will be strictly contained in another codimension 1 prime, which is a contradiction.

Similar arguments holds for $-D$ since we only need to first consider D and find the local defining function and take the reciprocal.

Then assume that in general we have some $\sum_i a_i [Y_i]$, then again we consider the union, and then the complement. On the complement, the divisor is defined by 1.

Then for each point y on the union of these codimension 1 subsets Y_i , we take the local ring at that point and in this local ring, we know that the generic points y_i of Y_i which y belongs to correspond to a codimension 1 prime, which is going to be principal, then we can assume that these functions are defined in the local ring at y , then defined on a small affine open around y . Then by shrinking this affine open around y as above, we can assume that each of the functions has exactly an order one zero at y_i , and no other zeros, clearly no poles. Then we know that by taking appropriate products of these functions, we recover D in this affine open, we are done. $\square$

Summarizing what we did above, we can now get a handle of the Picard group of a normal Noetherian scheme. We define the class group of X , denoted by $\text{Cl}X$. This is defined to be the quotient $\text{Weil}X/\text{Prin}X$. Notice that we have a well defined map:

$$\text{div} : \text{Pic}X \rightarrow \text{Cl}X, \quad \mathcal{L} \mapsto \text{div}(s)$$

where s is any rational section of $\mathcal{L}$. This is well defined because if we consider two rational sections s, s' , then we know that the difference of $\text{div}(s) - \text{div}(s')$ is the divisor defined by a rational function on X , which is principal. This can be thought as coming from:

$$\begin{array}{ccccc} \{(\mathcal{L}, s)\} / \sim & \xrightarrow{\text{div}} & \text{LocPrin}X & \hookrightarrow & \text{Weil}X \\ \downarrow & & \downarrow & & \downarrow \\ \text{Pic}X & \xrightarrow{=} & \{\mathcal{L}\} / \simeq & \dashrightarrow & \text{LocPrin}X / \text{Prin}X \hookrightarrow \text{Cl}X = \text{LocPrin}X / \text{Prin}X \end{array}$$

where every morphism is a group homomorphism and the above div defined is the induced dotted map. You notice that this is in fact an isomorphism if we only consider it to be in $\text{LocPrin}X/\text{Prin}X$ with an inverse $D \mapsto \mathcal{O}_X(D)$.

Moreover, if X is factorial, $\text{LocPrin}X \hookrightarrow \text{Weil}X$ is actually an isomorphism. Therefore div is in fact an isomorphism between $\text{Pic}X$ and $\text{Cl}X$.

In particular, if A is a UFD, since we know that it's class group is 0, then the Picard group of $\text{Spec} A$ is trivial. For example, the Picard group of $\mathbb{A}_k^n$ is trivial.

Remark 3.30. This class group of X is more or less only a topological data concerning only the information in codimension 1 or less. For example, we know that if Z is an irreducible closed subset of codimension 2. Then the Weil divisors of $U = X \setminus Z$ is still those Weil divisors of X intersecting with U . Notice that taking out Z doesn't affect the rational functions on X , thus the principal divisors are still principal divisors of X intersecting with U . Therefore the class group remains the same.

Now let's see what happens if we take out some Z which is irreducible closed of codimension 1. In this case, the class group changes in a controllable way. Let $U = X \setminus Z$ then note that we have a short exact sequence:

$$0 \rightarrow \mathbb{Z} \xrightarrow{1 \mapsto [Z]} \text{Weil} X \xrightarrow{\cap^U} \text{Weil}(U) \rightarrow 0$$

Notice that if we mod the principal divisors, we have:

$$\mathbb{Z} \rightarrow \text{Cl} X \rightarrow \text{Cl} U \rightarrow 0$$

Here we may lose injectivity on the first map since there are codimension 1 irreducible closed subset Y such that $[Y]$ is not principal, but some $m[Y]$ is. To see an example, consider the cone:

$$X = \text{Spec} k[x, y, z]/(xy - z^2)$$

Notice that this is reduced regular in codimension 1 (in fact it's smooth except at the closed point (x, y, z) and recall smooth implies regular). In fact using Serre's $R_1 + S_2$ condition, we can even prove this is normal, therefore everything in our discussion before works. But I claim that if we let Y to be the closed subset defined by (x, z) . Notice that since the quotient is a dimension 1 space $\text{Spec} k[y]$, we know that (x, y) is indeed codimension 1. However, I claim that $[Y]$ is not principal and $2[Y]$ is principal.

In fact, let's compute the divisor of x . First, notice that as a global section, the function x can only have zeros, therefore we only need to worry about codimension 1 points in $V(x)$. Notice that $V(x) = \text{Spec} k[y, z]/(z^2)$. Notice that the only codimension 1 prime in this is (y) because we know that if you take the radical of this ring, it's just $k[y]$. Then we know that the minimal prime in this ring is 0, which has inverse image (z) in $k[y, z]/(z^2)$, therefore we know that the only codimension 1 prime that is relevant to x is (x, z) which is the preimage of (z) , and we know that when we localize at (x, z) , y is invertible, which means that y is invertible, which means that z is a uniformizer of the maximal ideal. Then since $x = \frac{z^2}{y}$, we know that:

$$\text{div} x = 2[Y]$$

However, I claim that $[Y]$ is not principal. Otherwise, suppose that some $f \in K(X)^\times$ makes:

$$\text{div}(f) = [Y] \Rightarrow \frac{2f}{x} \in (k[x, y, z]/(xy - z^2))^\times$$

Notice that units has to be outside (x, y, z) and indeed elements in k^* are all invertible, therefore we know that $f^2 = cx$. But we know that the fraction field is $k(\frac{z}{y}, y)$, then we know that in the function field, we have:

$$\left(\frac{f}{r}\right)^2 = cy$$

However, if we consider the valuation of both sides at (y) , the right hand side is 1, but the left hand side is a square, thus has to be even, a contradiction.

As some applications of the exact sequence:

$$\mathbb{Z} \rightarrow \text{Cl} X \rightarrow \text{Cl} U \rightarrow 0$$

First, we note that if U is an open subscheme of $\mathbb{A}_k^n$, then we know that its complement are just some union of some irreducible closed subsets of various codimensions, at least 1. We know that taking out the codimension 2 or larger closed subsets doesn't change the class group, then we can assume that we only deal with codimension 1. Then we know that using induction, we know that the class group is still trivial. Therefore since U is factorial, then $\text{Pic}U = 0$ as well.

Another example could be take out the hypersurface in $\mathbb{P}_k^n$ determined by $x_0 = 0$, then we know that we have:

$$\mathbb{Z} \rightarrow \text{Cl}\mathbb{P}_k^n \rightarrow \text{Cl}\mathbb{A}_k^n \rightarrow 0$$

Therefore we know that the class group of $\mathbb{P}^n$ is generated by $[(x_0)]$. Then consider the isomorphism:

$$\text{Pic}\mathbb{P}_k^n \xrightarrow{\text{div}} \text{Cl}\mathbb{P}_k^n$$

We know that since $\mathcal{O}(1)$ is mapped to $[(x_0)]$, we know that the Picard group of $\mathbb{P}_k^n$ is generated by $\mathcal{O}(1)$. We thus obtained another way to show this fundamental fact.

Finally, we mention how to twist a line bundle by a divisor D .

3.5 Bezout's Theorem

The full strength of Bezout's theorem requires the knowledge of Hilbert Polynomials and we won't state it here because we won't use it in the proof of Weil Conjectures. The main purpose is to state and prove a few results that is related to Bezout's theorem and is useful in their own.

First, it's particularly good to know the following formulas relating the partials of a homogeneous degree d polynomial $f(x_0, \dots, x_n)$ and the partial derivatives of $F(y_1, \dots, y_n)$ where $f(x_0, \dots, x_n) = x_0^d F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})$.

Recall that let k be any field and let $k(t)$ be the field of rational polynomials, if we consider it's subring $k[t]$ and the usual formal derivatives on $k[t]$, we know that it uniquely extends to a derivative on $k(t)$ and in $k(t)$ if we have $\frac{g(t)}{h(t)}$, to compute it's derivative, we can use the formal quotient rule. Similar argument holds for partials: for example we consider the x_0 partial on $k[x_0, \dots, x_n]$, then it uniquely extends to the ring where we invert x_0 . This is basically the unique extension property of a derivation.

Then we know that if we compute in the localization and when $i \neq 0$:

$$\frac{\partial f(x_0, \dots, x_n)}{\partial x_i} = \frac{\partial}{\partial x_i} (x_0^d F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})) = x_0^d \frac{\partial F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})}{\partial \frac{x_i}{x_0}} \cdot \frac{1}{x_0} = x_0^{d-1} \frac{\partial F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})}{\partial \frac{x_i}{x_0}}$$

However, if we compute x_0 , we have:

$$\frac{\partial f(x_0, \dots, x_n)}{\partial x_0} = d x_0^{d-1} F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}) - x_0^{d-1} \sum_{i=1}^n \frac{\partial F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})}{\partial \frac{x_i}{x_0}} \frac{x_i}{x_0} = x_0^{d-1} (d F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}) - \sum_{i=1}^n x_i \frac{\partial F(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0})}{\partial \frac{x_i}{x_0}})$$

Using these formula, when you evaluate at a certain $(a_0, \dots, a_n)$, you also need to plug in $x_i = a_i$. Then let's use this to do a few useful examples which is related to special cases of Bezout's Theorem in general.

Example 3.31. Suppose that X is a degree d hypersurface in $\mathbb{P}_k^n$ cut out by some homogeneous polynomial of degree d , not necessarily irreducible. Recall that a line ℓ in $\mathbb{P}_k^n$ is the closed subscheme defined by the ideal generated by $n-1$ linearly independent linear forms in $k[x_0, \dots, x_n]$. Then we will make sense of what does it mean that the intersection of the line ℓ , which is not contained in X , and X contains d points with multiplicity. In particular, k doesn't have to be algebraically closed. *** to be completed

Example 3.32. Again, suppose that X is a degree d hypersurface cut out by a degree d homogeneous polynomial f . In the previous example, we made sense of what does the intersection multiplicity mean. In this example, suppose that $p \in X$ is a regular point of X which is also an intersection point with ℓ , then ℓ is contained in the tangent hypersurface of X . *** to be completed

Now suppose that k is a field, possibly finite, let $\{p_1, \dots, p_m\} \in \mathbb{P}_k^n$ be a collection of points not necessarily closed but none of which is the generic point of $\mathbb{P}_k^n$. Then we will show that for large enough N , there is a nonempty hypersurface X of degree N containing any of these p_i .

First, suppose that we only have one point p_1 . We need to find a hypersurface that doesn't contain p_1 .

*** to be completed

Now suppose that X is a closed subset of $\mathbb{P}_k^n$ with dimension at least 1 and H is a nonempty hypersurface in $\mathbb{P}_k^n$. Then we will first show that $H \cap X$ is never empty.

Indeed, suppose that H is defined by some degree d homogeneous polynomial f if we think about the affine cone of H , which is just:

$$\text{Spec}k[x_0, \dots, x_n]/f$$

Note that since f is homogeneous, then $f \in (x_0, \dots, x_n)$ which means that the origin is in the subscheme defined by f . Then consider the canonical map from the affine cone to the projective space:

$$\text{Spec}k[x_0, \dots, x_n]/f \setminus \{(0, \dots, 0)\} \rightarrow \text{Proj}k[x_0, \dots, x_n]/(f)$$

We can assume that X is irreducible and of dimension at least 1, which correspond to a homogeneous prime ideal $\mathfrak{p}$ in $k[x_0, \dots, x_n]$. Let's also consider the cone of $\mathfrak{p}$ which is $\text{Spec}k[x_0, \dots, x_n]/\mathfrak{p}$. Let's assume that in the projective space X is not contained in H which is equivalent to say that $f \notin \mathfrak{p}$. Therefore we know that f is going to be a nonzero divisor in $k[x_0, \dots, x_n]/\mathfrak{p}$, which means that $V(f)$ in $k[x_0, \dots, x_n]/\mathfrak{p}$ has codimension 1. Since X defined by $\mathfrak{p}$ in $\mathbb{P}_k^n$ has dimension at least 1, we know that $\mathfrak{p}$ is not a maximal ideal. Therefore we know that the dimension of $k[x_0, \dots, x_n]/\mathfrak{p}$ has dimension at least 1, therefore $V(f)$ can not be empty since each irreducible component of $V(f)$ has codimension 1 and we know that codimension 1 space in a dimension at least 1 space is not empty. Moreover, this intersection point is not the original in the affine cone since $\mathfrak{p}$ is a prime ideal that doesn't contain $(x_0, \dots, x_n)$, therefore the image of the intersection point in the affine cone is the intersection point of X and H in the projective space.

References

- [Mus11] Mircea Mustaă, *Zeta functions in algebraic geometry*, https://dept.math.lsa.umich.edu/~mmustata/zeta_book.pdf, 2011, Lecture notes for Math 732, University of Michigan.
- [Vak25] Ravi Vakil, *The rising sea: Foundations of algebraic geometry*, Princeton University Press, 2025.